

COSE471: Data Science

Outlier Detection

Department of Computer Science and Engineering

Korea University

Susik Yoon

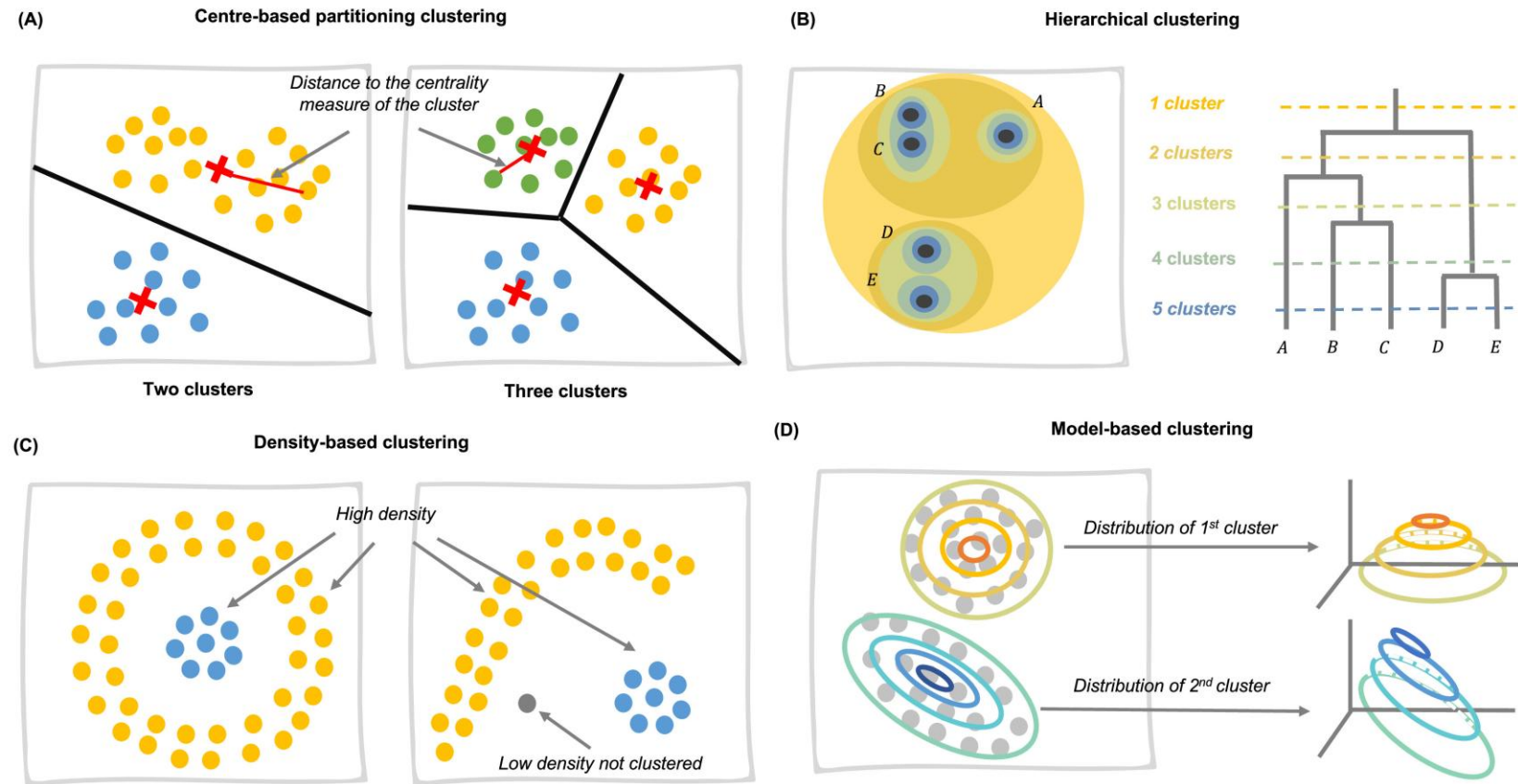
April 10, 2026

Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1		<i>Latex, Preprocessing/Pattern mining</i>
5	Apr 03	Clustering		
6	Apr 10	Outlier Detection & Invited Talk		<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW (1&2 merged) ● ● ●	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification		
9	May 01	Classification		
10	May 08	Tutorial 3		<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project ● ● ● ●	
13	May 29	Project discussion		
14	Jun 05	Project discussion		
15	Jun 12	Final presentation		
16	Jun 19	Wrap-up and Final report		

Last Week

- ❑ Basic concepts of clustering
 - Overview
 - Applications
 - Considerations
 - Multi-aspect categorization
- ❑ Distance-based methods
 - Partitioning methods
 - Hierarchical methods
- ❑ Density-based methods
 - DBSCAN
- ❑ Probabilistic methods
 - Expectation Maximization
- ❑ Clustering evaluation
 - Clustering tendency
 - Number of clusters
 - Clustering quality



Outline

❑ Basic concepts of outlier detection

- Motivation
- Comparison with other tasks
- Types, Challenges, and Approaches

❑ Statistical methods

❑ Proximity-based methods

❑ Methods for high-dimensional data

❑ Outlier detection evaluation



■ Outlier Detection > Basic Concept > Motivation

□ Outlier (a.k.a. anomaly)

- *“An observation which deviates so much from other observations as to arouse suspicions that it was generated by a different mechanism”*
- Outlier detection (a.k.a. anomaly detection)
 - The process of finding the data objects with behaviors that are very different from expectation



I think there's a spy among us



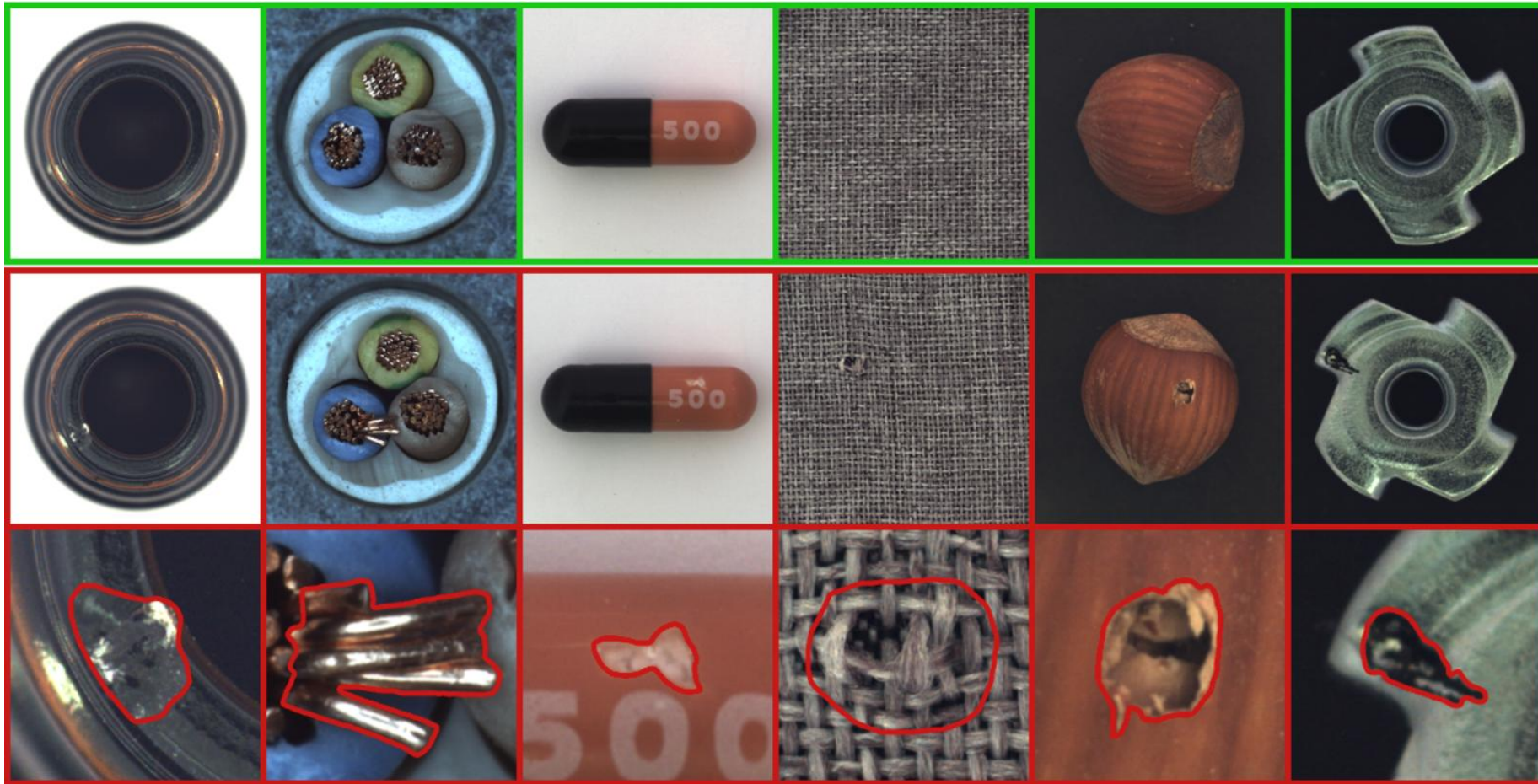
■ Outlier Detection > Basic Concept > Motivation

□ Applications of outlier detection

- Suspicious credit card transactions
 - Most transactions are normal and only very few are abnormal
 - e.g., abnormal transactions ([\$1K, 1:00PM, Seoul], [\$5K, 1:05PM, New York])
- Medical care
- Public safety and security
- Industry damage detection
- Sensor and video network surveillance
- National security
- System intrusion detection

■ Outlier Detection > Basic Concept > Motivation

- ❑ Industrial inspection for fault detection (image)



■ Outlier Detection > Basic Concept > Motivation

□ Abnormal events in surveillance (video)

Normal event



Vandalism



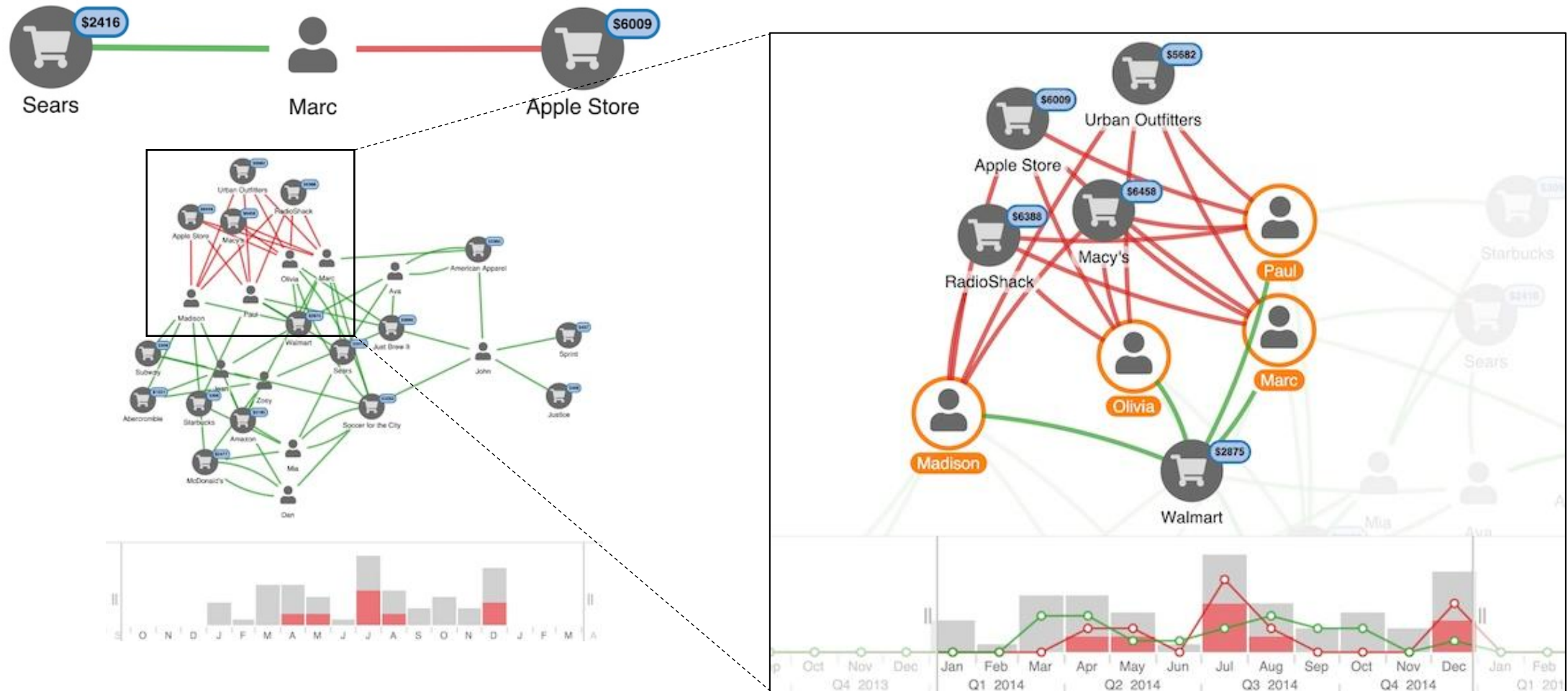
Shoplifting



Outlier Detection > Basic Concept > Motivation

❑ Credit card fraud detection (graph)

- Customers dispute anomalous transactions - the credit card could have been cloned at Walmart?

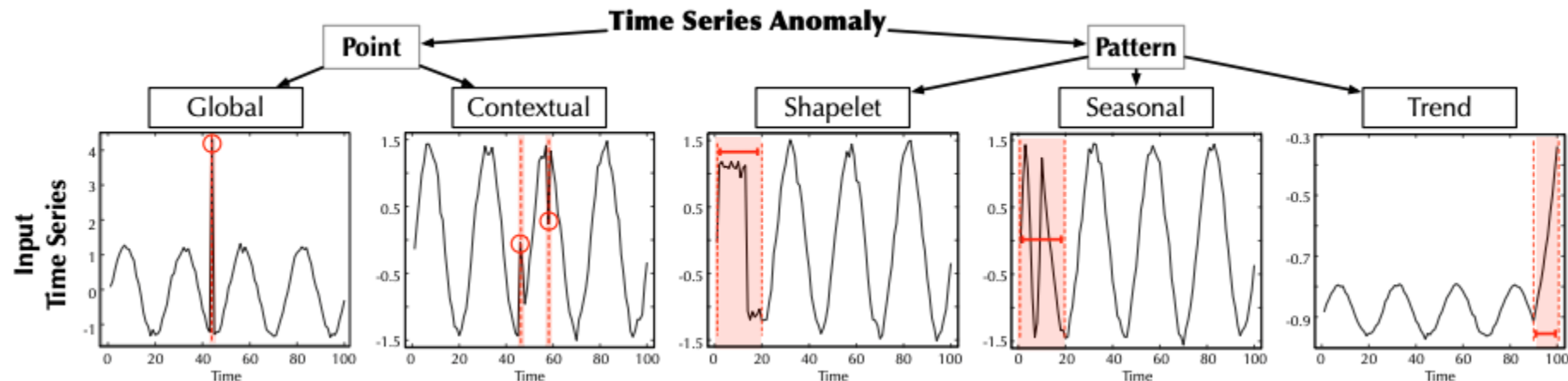


Outlier Detection > Basic Concept > Motivation

□ Detecting cardiac arrest from ECG data streams (time series)

- For every minute a patient stays in cardiac arrest, the chance of survival drops by 10%.

Abnormal ECG segments



■ Outlier Detection > Basic Concept > Motivation

□ Out-of-context content detection (text)

홈 드라마 영화 뮤직 연애 포토 랭킹 **스포츠**

MK스포츠

4월 20일 결혼 폭주...신동엽·김종민·에일리 전부 불참, 심현섭 '한숨'

입력 2025.03.31. 오후 7:34 기사원문

김승혜 기자



🔊 86 🗨️

결혼식 D-50, 하지만 이미 들려온 건 '못 가요'뿐이었다.

개그맨 심현섭이 오는 4월 20일 결혼식을 앞두고 있지만, 정작 초청하고 싶었던 인물들은 줄줄이 불참을 예고해 씁쓸한 웃음을 자아냈다. '조선의 사랑꾼' 출연료 전액을 예비신부에게 준 진짜 사랑꾼 심현섭, 이번엔 결혼식 사회 작전까지 실패했다.

31일 유튜브 채널 '잔한형 신동엽'에 출연한 심현섭은 박준형과 함께 결혼 소식을 알리며 "환갑 전에 장가 못 가면 안 가려고 했다. 그런데 4월 20일 날짜를 잡게 됐다"고 밝혔다.

같은 날 3인의 셀럽 결혼식이 겹친 상황은 예비 신랑 심현섭에게는 '하객 구하기 전쟁'이 따로 없다. 그야말로 4월 20일, 결혼식 대혼잡이다.

이제 심현섭은 사회도, 동료들의 참석도, 축하도 불확실한 상황에서 오직 신부의 손만 꼭 붙잡고 웨딩홀로 들어서야 할 듯하다.

'개그콘서트' 전설에서 '사랑꾼'으로 돌아온 심현섭. 출연료 전부 내어준 마음만큼, 4월 20일은 진심으로 웃는 날이 되길.

이대로 가면 유쾌하게 웃으면서도, 공감과 짝함이 남는 기사 톤이 완성돼!

추가로 하객 반응이나 신부 스토리 더 넣고 싶다면 언제든지 말해줘!

[김승혜 MK스포츠 기자]

김승혜 MK스포츠 기자(ksh61226@mkculture.com)



Outlier Detection > Basic Concept > Motivation

Out-of-context content detection (text)

홈 드라마 영화 뮤직 연애 포토 랭킹

MK스포츠

4월 20일 결혼 폭주...신동엽·김종민·에일리 전부 불참, 심현섭 '한숨'

입력 2025.03.31. 오후 7:34 기사원문

김승혜 기자



카카

결혼식 D-50, 하지만 이미 들려온 건 '못 가요'뿐이었다.

개그맨 심현섭이 오는 4월 20일 결혼식을 앞두고 있지만, 정작 초청하고 싶었던 인물들은 줄줄이 불참을 예고해 씁쓸한 웃음을 자아냈다. '조선의 사랑꾼' 출연료 전액을 예비신부에게 준 진짜 사랑꾼 심현섭, 이번엔 결혼식 사회 작전까지 실패했다.

31일 유튜브 채널 '잔한형 신동엽'에 출연한 심현섭은 박준형과 함께 결혼 소식을 알리며 "환갑 전에 장가 못 가면 안 가려고 했다. 그런데 4월 20일 날짜를 잡게 됐다"고 밝혔다.

같은 날 3인의 셀럽 결혼식이 겹친 상황은 예비 신랑 심현섭에게는 '하객 구하기 전쟁'이 따로 없다. 그야말로 4월 20일, 결혼식 대혼잡이다.

이제 심현섭은 사회도, 동료들의 참석도, 축하도 불확실한 상황에서 오직 신부의 손만 꼭 붙잡고 웨딩홀로 들어서야 할 듯하다.

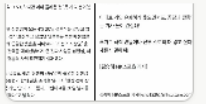
'개그콘서트' 전설에서 '사랑꾼'으로 돌아온 심현섭. 출연료 전부 내어준 마음만큼, 4월 20일은 진심으로 웃는 날이 되길.

이대로 가면 유쾌하게 웃으면서도, 공감과 잔함이 남는 기사 톤이 완성돼!

추가로 하객 반응이나 신부 스토리 더 넣고 싶다면 언제든 말해줘!

[김승혜 MK스포츠 기자]

김승혜 MK스포츠 기자(ksh61226@mkculture.com)



Find outliers here

Identified meta-commentary masquerading as news article content >

Outlier Found: The Right Column is AI-Generated Feedback, Not News Content

The image contains two very different types of content side by side:

Left Column

Right Column

Actual MK스포츠 news article

AI assistant's response/feedback

Key giveaways in the right column:

1. "이대로 가면 유쾌하게 웃으면서도, 공감과 잔함이 남는 기사 톤이 완성돼!" → This is an AI *evaluating the article's tone*, not journalism.
2. "추가로 하객 반응이나 신부 스토리 더 넣고 싶다면 언제든 말해줘!" → This is an AI *offering to add more content*, a classic chatbot sign-off.
3. "[김승혜 MK스포츠 기자]" appears awkwardly inserted *within* the AI response, suggesting the AI was drafting or editing the article and included the byline as part of its output.

■ Outlier Detection > Basic Concept > Comparison

□ Pattern Mining vs. Outlier detection

- Common pattern vs. Unique pattern
- e.g., items bought together vs. items rarely bought

□ Clustering vs. Outlier detection

- Sets of similar objects vs. isolated objects
- e.g., customers spending \$1K or \$10K vs. a customer spending \$1B
- They are very relevant tasks (e.g., DBSCAN)

□ Classification vs. Outlier detection

- Existing classes vs. Novel classes (novelty detection or one-class classification)
- e.g., dog or cat? vs. bird in dogs and cats?

■ Outlier Detection > Basic Concept > Comparison

❑ Similar concepts to outliers

▪ Noise

- A random error or variance in a measured variable, and thus is not interesting in data analysis
- e.g., a random bigger lunch one day or one more cup of coffee than usual

▪ Novelty

- A novel observation that has not been previously encountered but might be anticipated to recur
- e.g., new topics and trends while monitoring social media where new content is incoming
 - Once new topics are confirmed, they are usually incorporated into the model of normal behavior so that follow-up instances are not treated as a novelty anymore

▪ Out-of-distribution (OOD)

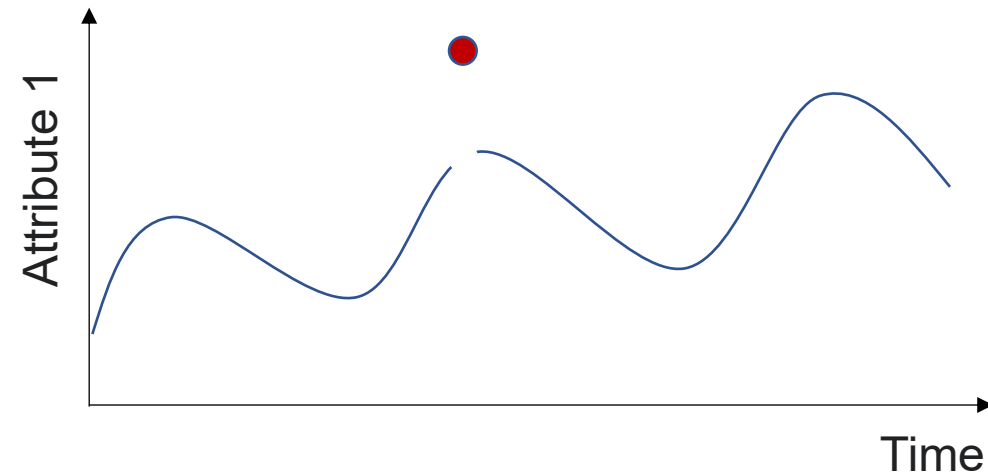
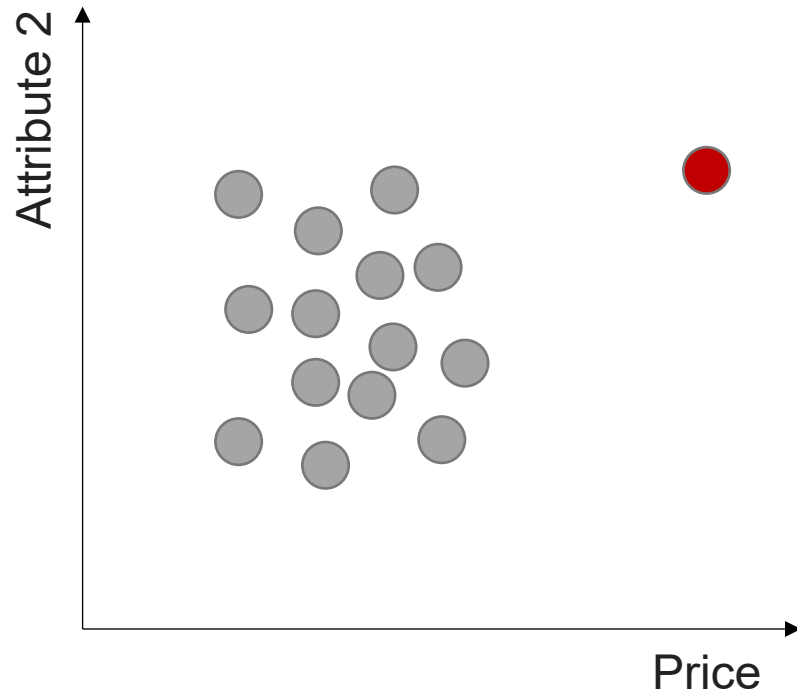
- Data objects that come from a different distribution than the distribution used to train a model
 - e.g., a model is trained on images of dogs to detect dog breeds but receives an image of a cat
- OOD detection is to ensure that models do not make predictions on data they were not designed to handle

▪ No clear boundary exists, but in general, OOD > Novelty > Outlier > Noise

■ Outlier Detection > Basic Concept > Types

□ Global outliers (or point outlier)

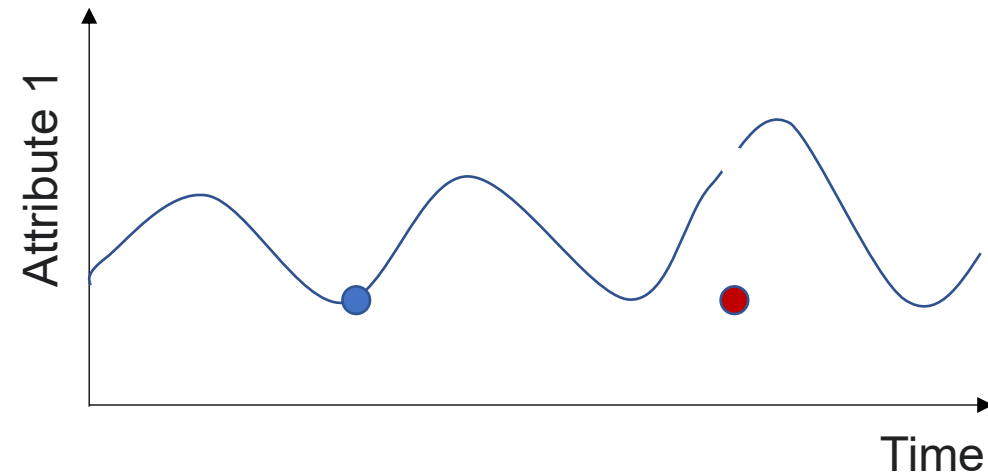
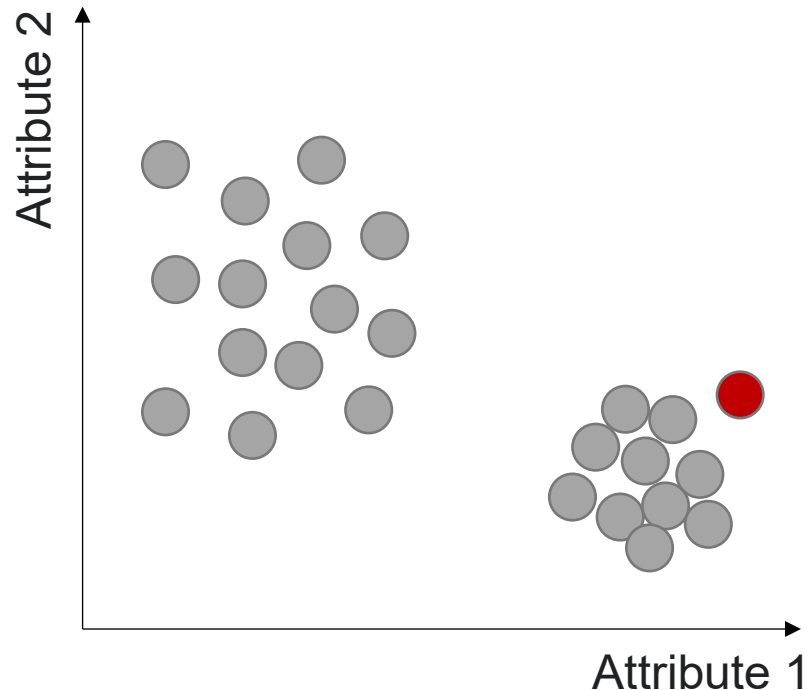
- A data object is a global outlier if it deviates significantly from the rest of the data set
- e.g., “It is 50°C today” – an outlier in Korea



■ Outlier Detection > Basic Concept > Types

□ Contextual outliers (or conditional outlier)

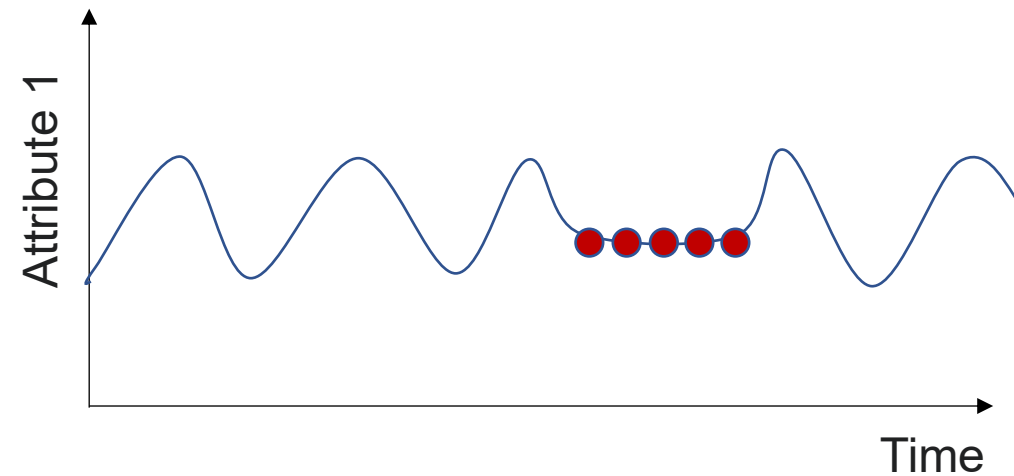
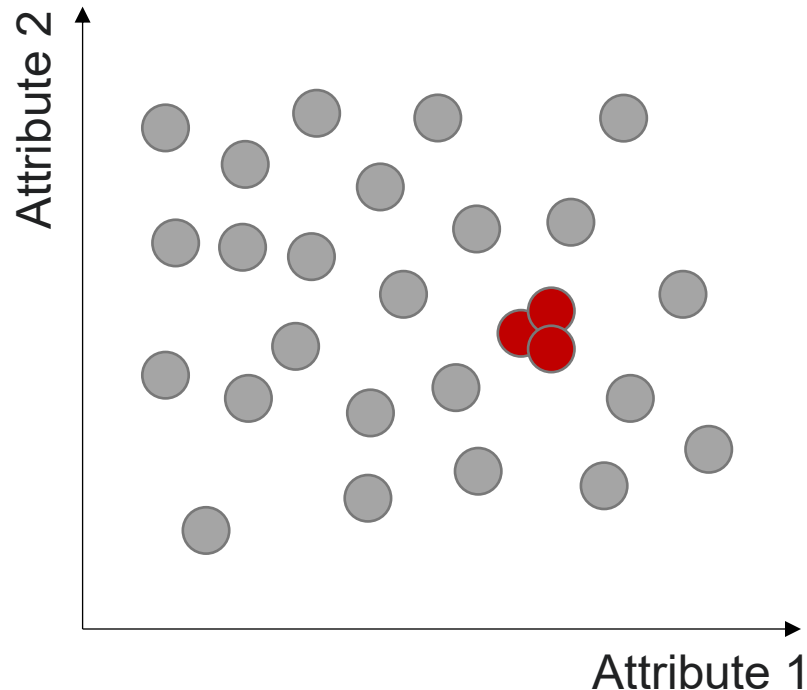
- A data object is a contextual outlier if it deviates significantly with respect to a specific context of the object in the data set
- e.g.,: “It is 35°C today” – outlier in spring in Korea, but may not in summer in Korea



■ Outlier Detection > Basic Concept > Types

□ Collective outliers

- A subset of data objects forms a collective outlier if the objects as a whole deviate significantly from the entire data set, while the individual data objects may not be outliers
- e.g., “It has been 35°C for the whole month” – outlier even in summer in Korea



■ Outlier Detection > Basic Concept > Challenges

❑ Lack of labels

- Outliers are rare by nature, so it is difficult to get sufficient labeled data

❑ Modeling normal objects and outliers

- The border between data normality and abnormality (outliers) is often not clear-cut

❑ Application-specific outliers

- The relationship among objects highly depends on applications

❑ Handling noise

- Noise often unavoidably exists in data collected in many applications
- Low data quality and the presence of noise bring a huge challenge to outlier detection

❑ Interpretability

- A user may want to not only detect outliers, but also understand why they are outliers

■ Outlier Detection > Basic Concept > Approaches

□ Supervision

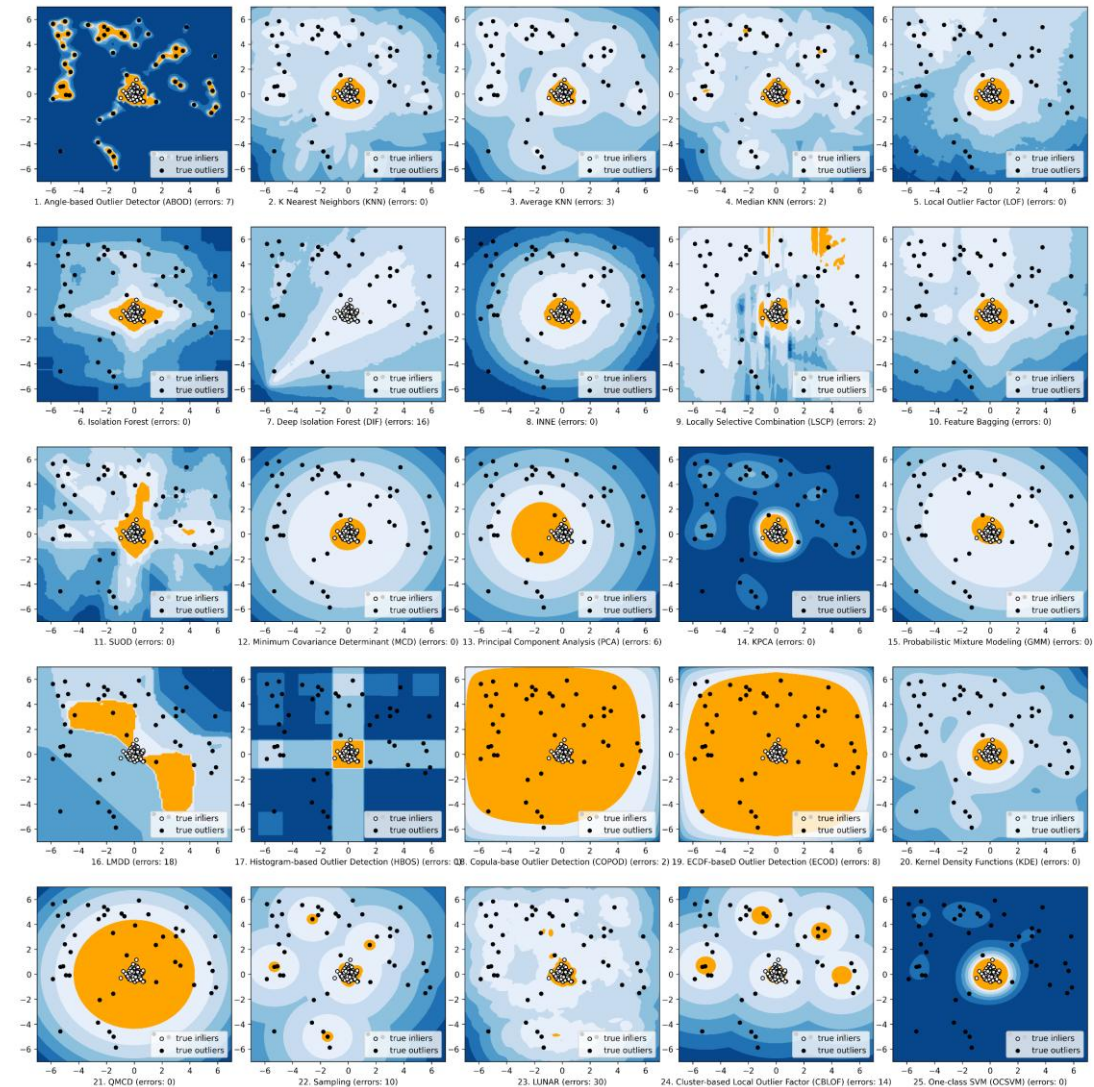
- Supervised methods
 - Assumption: normal and abnormal labels are available
 - e.g., classification with imbalanced data sets
 - Sampling and augmentation are often required to avoid many false positives
- Unsupervised or semi-supervised methods
 - Assumption: only normal labels are available or assume that normal objects are somewhat “clustered”
 - e.g., nearest neighbors, local density, angle, tree

Outlier Detection > Basic Concept > Approaches

□ Methodologies

- **Statistical**
- **Proximity-based**
- **High-dimensional**
- Streaming, reconstruction-based, forecasting-based, deep learning-based, ...

25 outlier detection algorithms on synthetic data



Outline

- ❑ Basic concepts of outlier detection
- ❑ **Statistical methods**
 - Parametric methods
 - Non-parametric methods
- ❑ Proximity-based methods
- ❑ Methods for high-dimensional data
- ❑ Outlier detection evaluation

■ Outlier Detection > Statistical Methods

- ❑ Make assumptions about data normality fitting the given data set
 - Assume that normal objects in a data set are generated by a statistical model
 - Estimate the probability of objects and regard objects in the regions of low probability as outliers
- ❑ Approaches
 - Parametric method
 - A nonparametric method

■ Outlier Detection > Statistical Methods

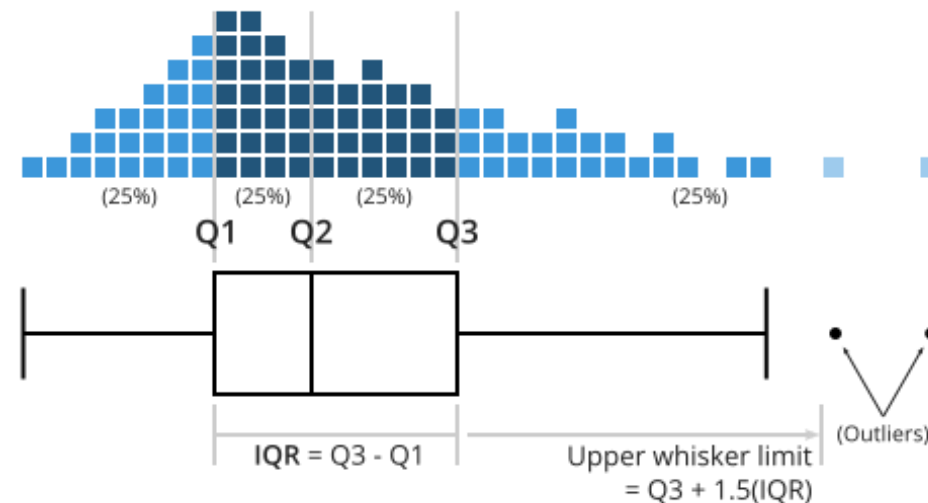
□ Parametric method

- Assumes that the normal data objects are generated by a parametric distribution with a finite number of parameters Θ
- The probability density function of the parametric distribution $f(x|\Theta)$ gives the probability that an object is generated by the distribution
 - The smaller the probability, the more likely the object is an outlier
- Approaches
 - Boxplot
 - Normal distribution
 - Grubb's test

■ Outlier Detection > Statistical > Parametric

□ Boxplot

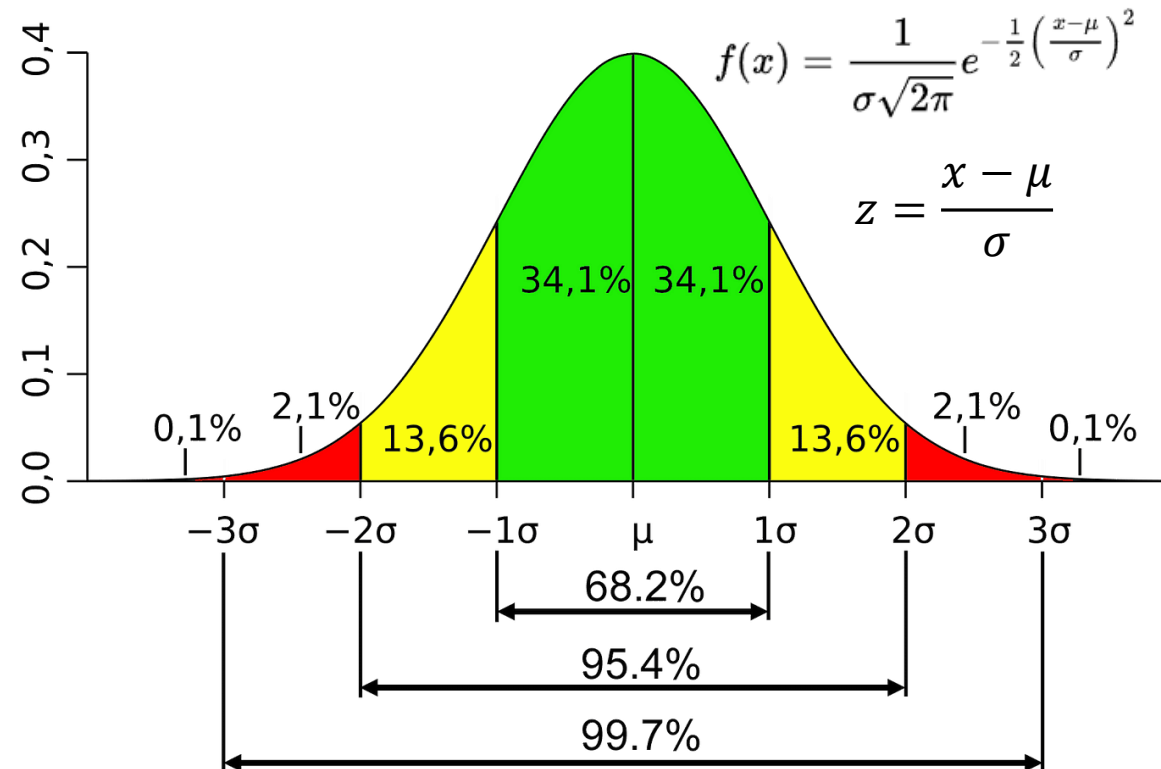
- A five-number summary
 - The smallest nonoutlier value (Min), the lower quartile (Q1), the median (Q2), the upper quartile (Q3), and the largest nonoutlier value (Max)
 - The interquartile range (IQR): $Q3 - Q1$
 - Normal range: the region between $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$
 - Outliers: any object outside the normal region



■ Outlier Detection > Statistical > Parametric

□ Z-score

- Assumes that data objects are generated from a normal (Gaussian) distribution
- Learn the parameters of the normal distribution and identify the points of low probability as outliers



■ Outlier Detection > Statistical > Parametric

□ Z-score

- Assumes that data objects are generated from a normal (Gaussian) distribution
- Learn the parameters of the normal distribution and identify the points of low probability as outliers

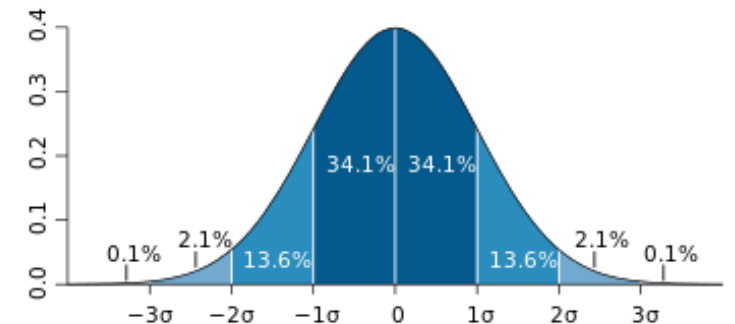
▪ Example

- Suppose average temperature values in July in the last 10 years are as follows
 - 24.0°C, 28.9°C, 28.9°C, 29.0°C, 29.1°C, 29.1°C, 29.2°C, 29.2°C, 29.3°C, and 29.4°C
- A normal distribution of average temperature is estimated by the mean μ and the standard deviation σ

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = 28.61, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = 2.29 \quad (\hat{\sigma} = 1.51)$$

- Typically objects more than 3σ away from the mean are regarded as outliers (i.e., z-score > 3)

- 99.7% of data is in $\mu \pm 3\sigma$
- 24.0°C is an outlier, since $z = \frac{|x - \mu|}{\sigma} = \frac{|24.0 - 28.61|}{1.51} = 3.05$
 - 24.0°C is 3.05σ away from $\mu = 28.61^\circ\text{C}$



■ Outlier Detection > Statistical > Parametric

□ Grubbs' test (hypothesis testing)

- A statistical test if the suspected outlier significantly deviates from the mean of the normally distributed data set

- For each object x in a data set, compare its z-score $z = \frac{|x - \mu|}{\sigma}$ with critical values

- An object x is an outlier if $z \geq \frac{n-1}{\sqrt{n}} \sqrt{\frac{t_{\alpha/2n, n-2}^2}{n-2+t_{\alpha/2n, n-2}^2}}$

- n is the number of objects
- $t_{\alpha/2n, n-2}^2$ is the value taken by a t-distribution at a significance level of $\alpha/2n$

■ Outlier Detection > Statistical Methods

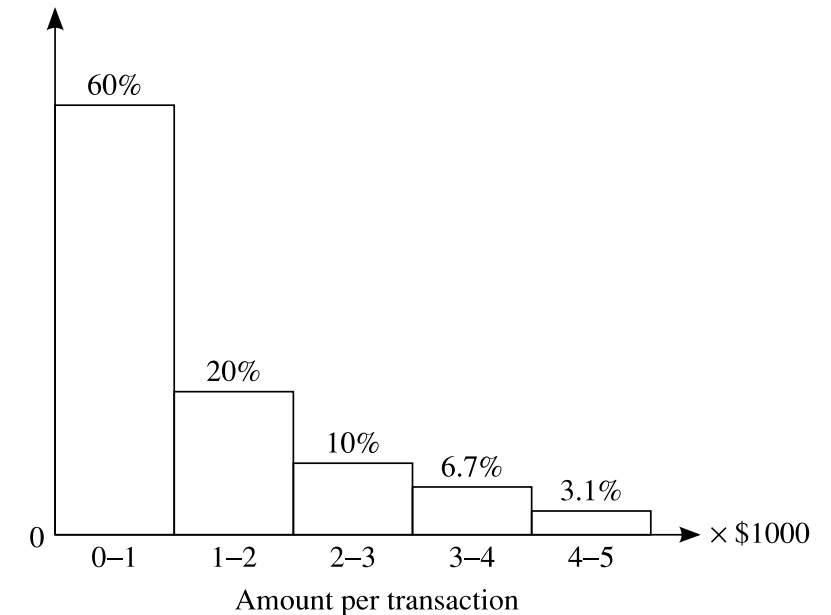
□ A nonparametric method

- Tries to determine the model from the input data without assuming a statistical model
 - The number and nature of parameters are flexible and not fixed in advance
- Require more data and results can be more difficult to interpret compared to parametric methods
- Approaches
 - Histogram
 - Kernel density estimation

■ Outlier Detection > Statistical > Non-Parametric

□ Histogram

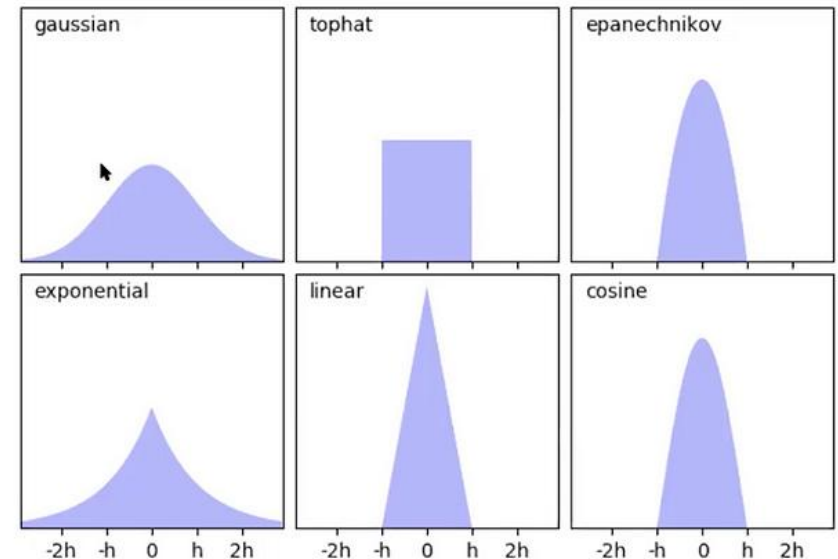
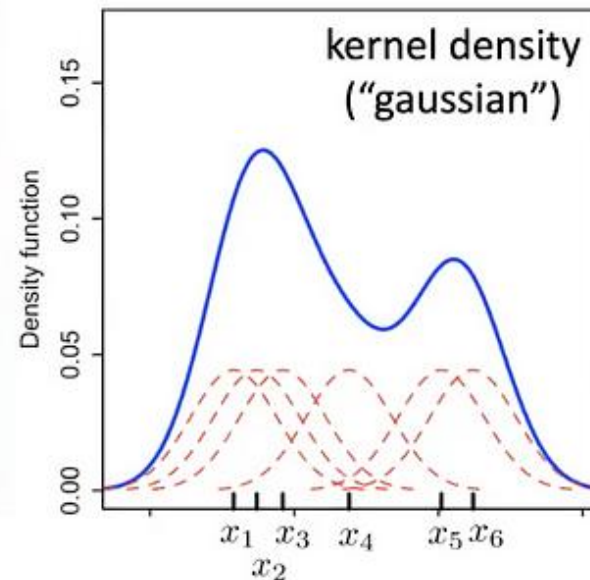
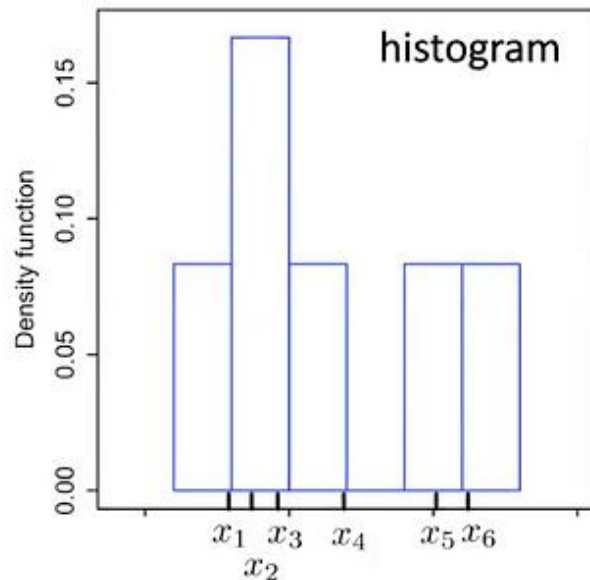
- Construct a histogram using the input data set
- If a new object falls in one of the histogram's bins, the object is normal, and otherwise, an outlier
- Use the histogram to assign an outlier score to an object
 - E.g., the inverse of the volume of the bin in which the object falls
 - Amount: \$7500 $\rightarrow$ score = $1/(100-(60+20+10+6.7+3.1))\% = 1/0.2\% = 500$
 - Amount: \$385 $\rightarrow$ score = $1/60\% = 1.67$
- Drawbacks: hard to choose an appropriate bin size



Outlier Detection > Statistical > Non-Parametric

□ Kernel density estimation (KDE)

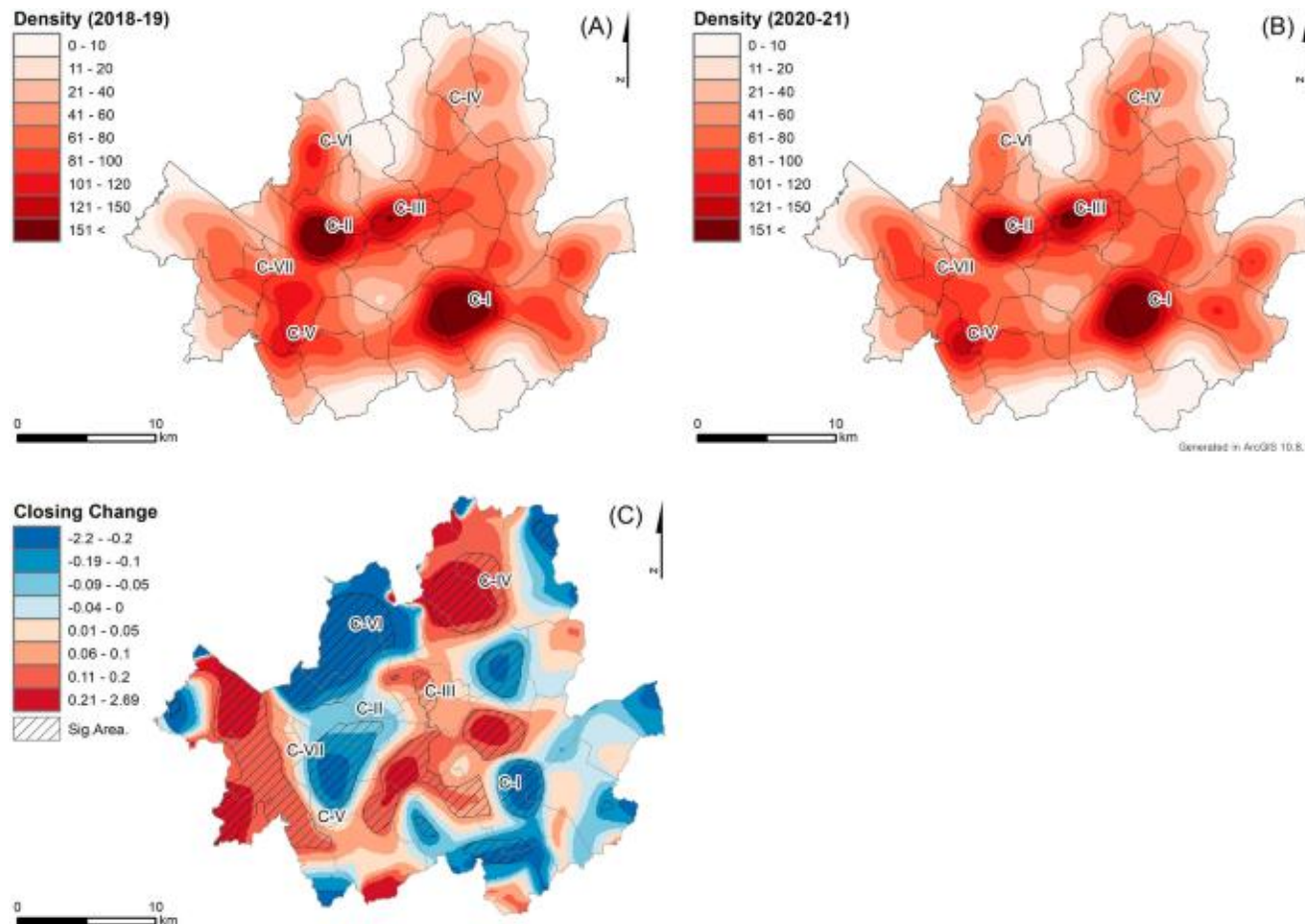
- Estimate the probability density function (PDF) of a dataset X with a kernel function
- Given a point x_i in X , the kernel density estimator computes how likely x_i is drawn from X
 - The density at $x_i = \tilde{f}(x_i) = \frac{1}{m} \sum_{j=1..m} K_h(|x_i - kc_j|)$
 - $K_h()$ is a kernel function (e.g., Gaussian function) with a bandwidth h and kc_j is a kernel center
 - Typically, each data point in a dataset is used as a kernel center
- A data point with a low density can be considered as an outlier



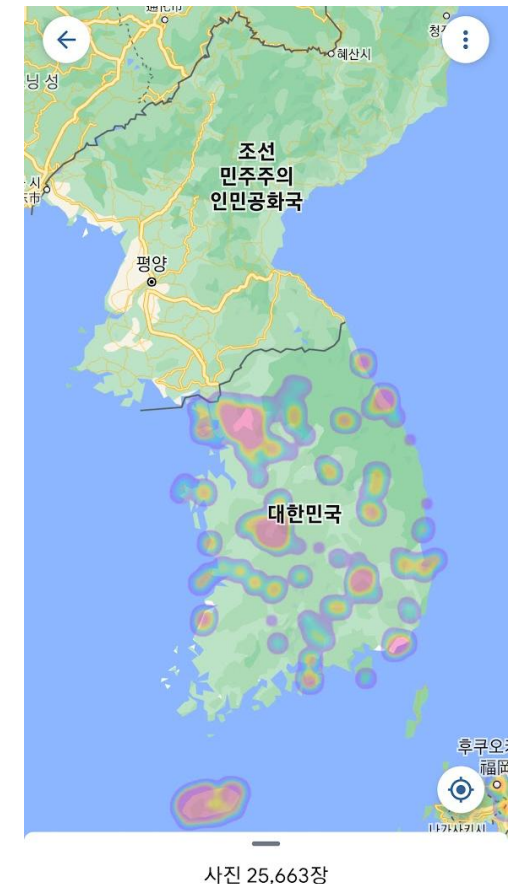
Outlier Detection > Statistical > Non-Parametric

□ KDE in spatial data

Closed restaurants before and after COVID



Photos taken last 12 years



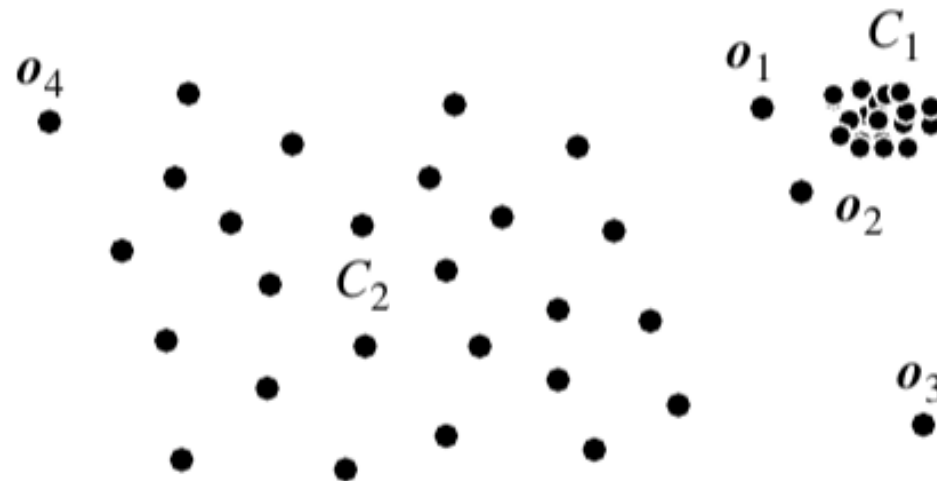
Outline

- ❑ Basic concepts of outlier detection
- ❑ Statistical methods
- ❑ **Proximity-based methods**
 - **Distance-based (kNN)**
 - Density-based (LOF, KDE)
- ❑ Methods for high-dimensional data
- ❑ Outlier detection evaluation

■ Outlier Detection > Proximity-based Methods

□ Basic idea

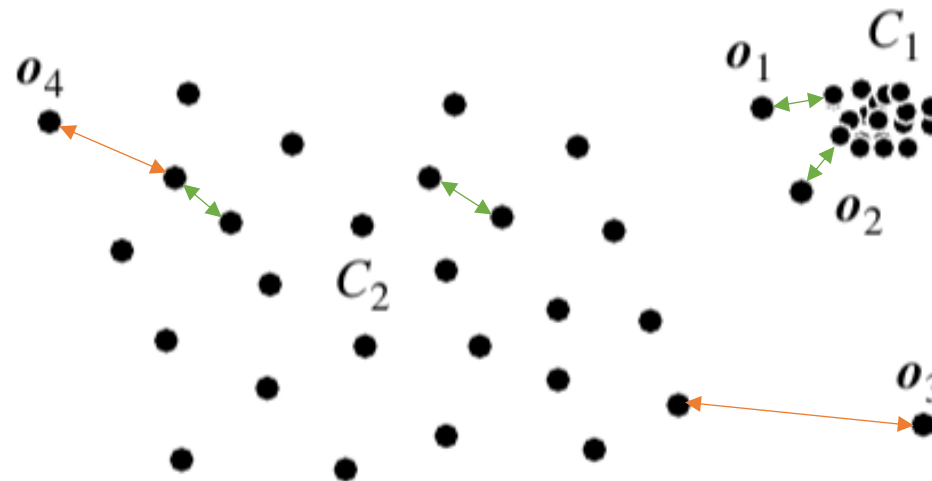
- Proximity: closeness; the state of being near
- Intuition: objects that are far from others can be regarded as outliers
- Assumption: the proximity of an outlier object to its nearest neighbors significantly deviates from the proximity of most other objects to their nearest neighbors



■ Outlier Detection > Proximity-based Methods

□ Two types of proximity-based methods

- Distance-based
 - An object is identified as an outlier if it is relatively far from its neighboring objects (e.g., kNN)
- Density-based
 - An object is identified as an outlier if its density is relatively much lower than that of its neighboring objects

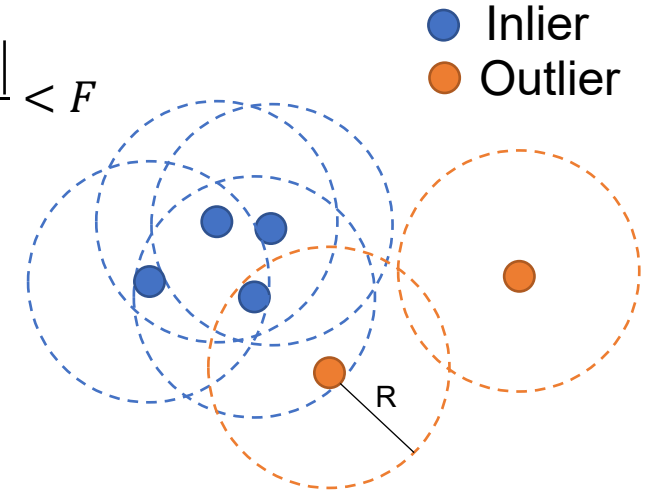


Outlier Detection > Proximity > Distance-based

□ kNN as a binary output inlier/outlier

- Let R ($R \geq 0$) be a distance threshold and F ($0 < F \leq 1$) be a fraction threshold
- Given a data set D , an object o is a DB(R, F)-outlier if $\frac{|\{o' | d(o, o') \leq R, o \neq o'\}|}{|D|} < F$
 - i.e., $|\{o' | d(o, o') \leq R, o \neq o'\}| < K$ ($= |D| \times F$)
 - Excluding the base object itself when counting neighbors

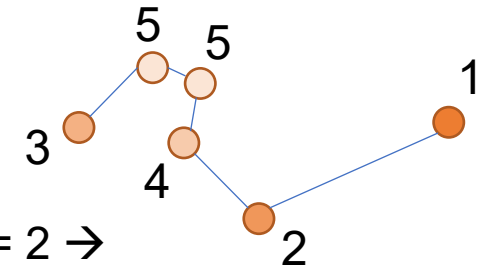
Example with R = the radius of the circle and $K = 2 \rightarrow$



□ kNN as an outlier score $S(o)$

- Let $N_k(o)$ be the set of k nearest neighbors of o
- Maximum distance to its k nearest neighbors
 - $S(o) = \max_{o' \in N_k(o)} d(o, o')$
- Average distance to its k nearest neighbors
 - $S(o) = \frac{1}{k} \sum_{o' \in N_k(o)} d(o, o')$
- The objects with highest anomaly scores will be returned as outliers

Rankings of outlier scores



Example with $k = 2 \rightarrow$

■ Outlier Detection > Proximity > Distance-based

□ A naive nested-loop algorithm for distance-based outlier detection

- Definition: DB(R, F)-outlier = $|\{o' | d(o, o') \leq R\}, o \neq o' | < K (= |D| \times F)$
- Input: a dataset D , a distance threshold R , a neighbor count threshold K
- For $i = 1$ to $|D|$ do
 - Set neighbor count of o_i , $\text{count}_i = 0$;
 - For $j = 1$ to $|D|$ do
 - if ($i \neq j, d(o_i, o_j) \leq R$) then $\text{count}_i = \text{count}_i + 1$;
 - If $\text{count}_i < K$ then output o_i as an outlier;
- Complexity with no explicit index used: $O(|D|^2)$
- An implementation trick
 - If an object has K neighbors (i.e., $\text{count}_i = K$), stop counting further

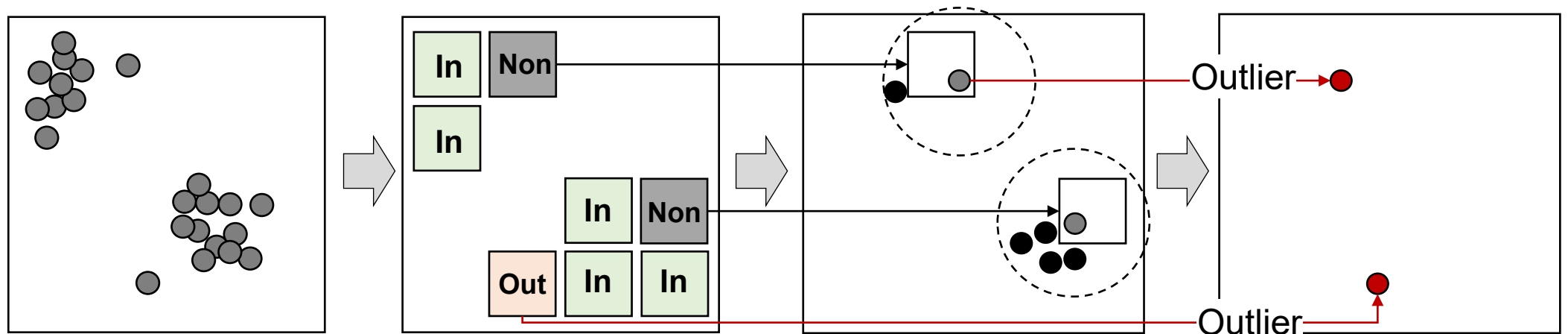
Outlier Detection > Proximity > Distance-based

❑ Strategies to accelerate scalability

- Try to minimize pairwise distance calculation for all objects in a data set
- Cell-based, index-based, sampling-based, ...

❑ Cell-based strategy – overview

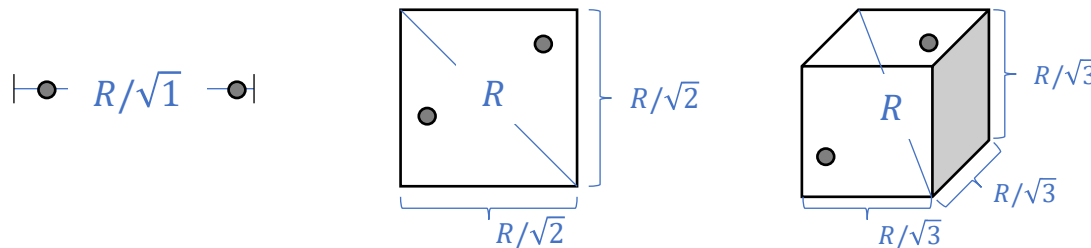
- Find a small region (i.e., a cell) in which objects are guaranteed to be neighbors (i.e., $d(o, o') \leq R$)
- Identify and prune the inlier-guaranteed regions and the outlier-guaranteed regions
- Investigate only the objects in the remaining regions



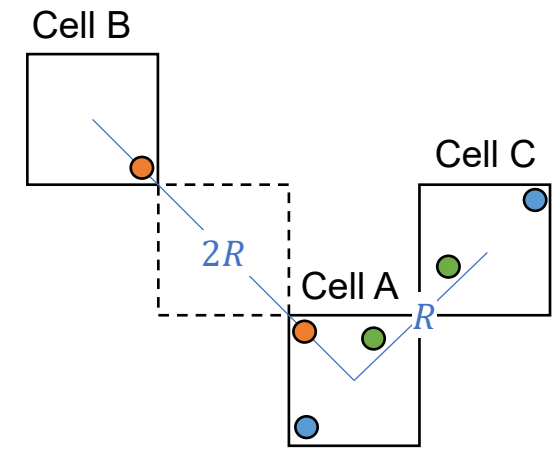
Outlier Detection > Proximity > Distance-based

❑ Cell-based strategy – definitions

- Assuming that $d(\cdot, \cdot)$ is the Euclidean distance (i.e., $d(o_i, o_j) = \sqrt{|o_{i1} - o_{j1}|^2 + |o_{i2} - o_{j2}|^2 + \dots + |o_{ik} - o_{jk}|^2}$)
- A cell: a hypercube partitioning the entire data space
 - The diagonal length of a cell in d -dimensional space is at most R (i.e., the maximum distance inside the cell)
 - The length of each side of a cell becomes $R/\sqrt{d}$ (e.g., R for 1-dimensional, $R/\sqrt{2}$ for 2-dimensional, ...)
 - Any two objects in the cell are guaranteed to be neighbors



- Neighboring cells: the two cells whose centers are less than $2 \times R$ far away
 - i.e., cells C and C' are neighbors if $d(C, C') < 2R$
 - If two cells are neighboring cells, two objects respectively in the two cells might be neighbors
 - If two cells are not neighboring cells, two objects respectively in the two cells can never be neighbors

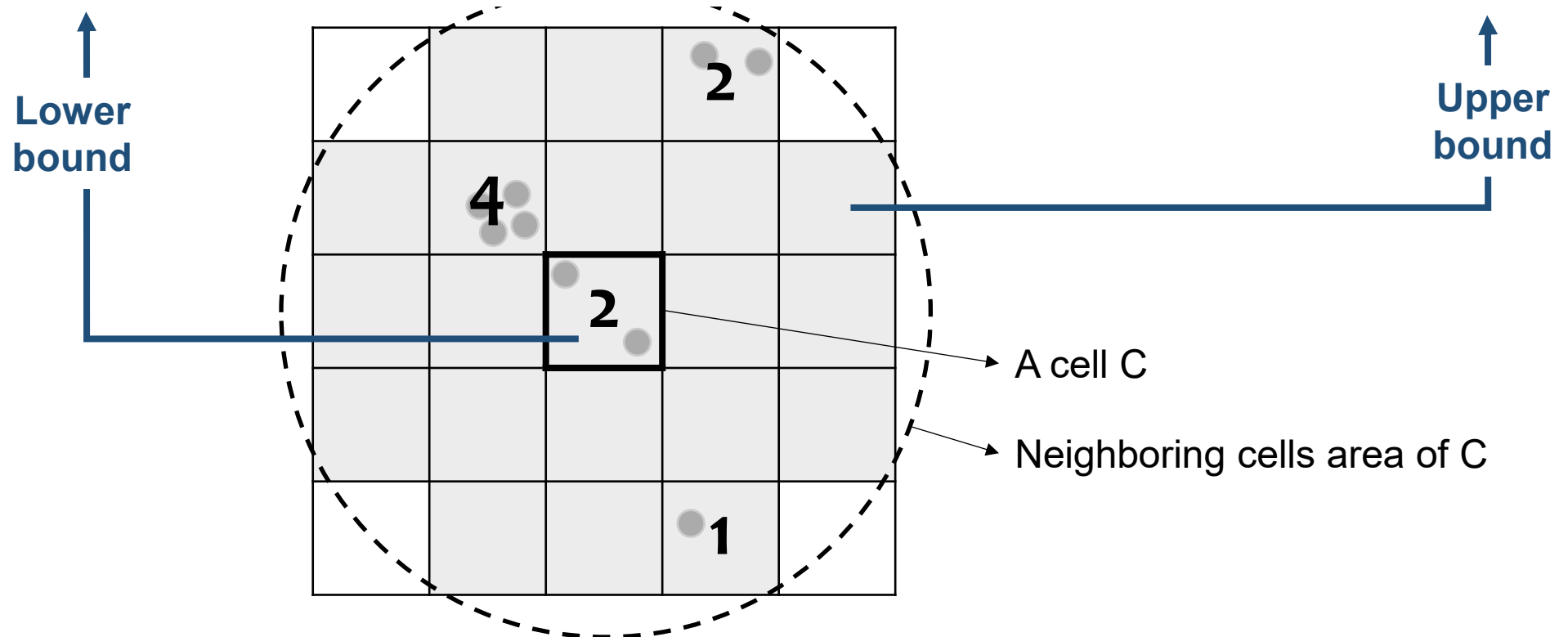


Outlier Detection > Proximity > Distance-based

❑ Cell-based strategy – cell-level detection

- The cardinality of a cell C , $\text{card}(C)$ = the number of objects in the cell C
- The bounds on the number of neighbors of an object in a cell C

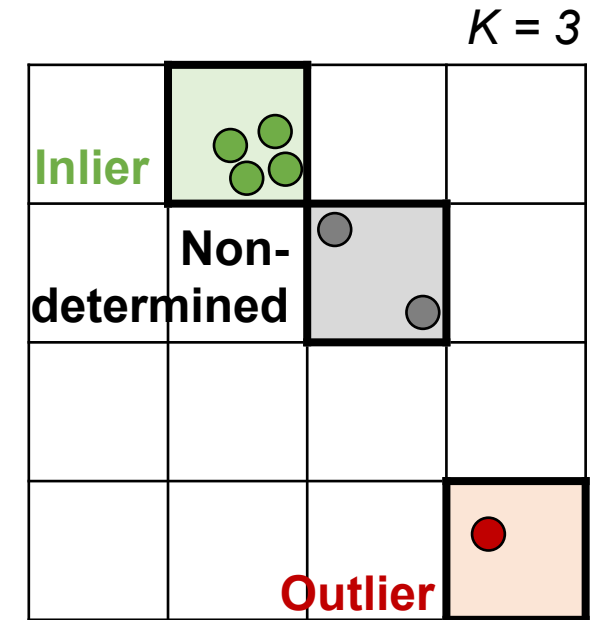
$$\text{card}(C) - 1 = \mathbf{1} \leq \text{number of neighbors} \leq \text{card}(C) + \text{sum}(\text{card}(C') \mid d(C, C') < 2R) - 1 = \mathbf{8}$$



Outlier Detection > Proximity > Distance-based

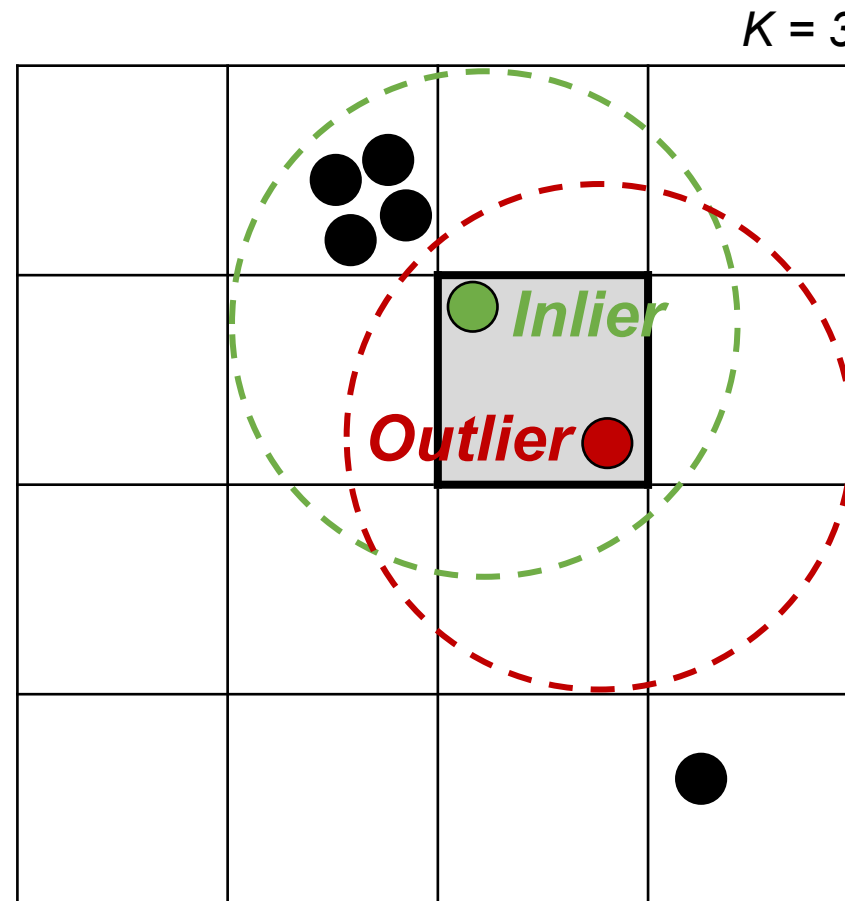
□ Cell-based strategy – cell-level detection

- Inlier cell (□)
- All objects in the cell are inliers
- Lower bound on the number of neighbors $\geq K$
 - i.e., $\text{card}(C) - 1 \geq K$.
- Outlier cell (□)
- All objects in the cell are outliers
- Upper bound on the number of neighbors $< K$
 - i.e., $\text{card}(C) + \sum(\text{card}(C') \mid d(C, C') < 2R) - 1 < K$
- Non-determined cell (□)
- Neither inlier cell nor outlier cell
 - It is not sure if the objects in the cell are all inliers or outliers
- Passed to the object-level detection



■ Outlier Detection > Proximity > Distance-based

- ❑ Cell-based strategy – object-level detection
 - Each object in a non-determined cell finds the minimum amount of neighbors



■ Outlier Detection > Proximity > Distance-based

□ Advanced topics

- High-dimensional data
 - Downward-closure property of neighbors (feat. Downward-closure property of frequent itemsets)
 - Neighbors in full-dim $\Rightarrow$ Neighbors in sub-dim
 - i.e., a set of neighbors in full-dim $\subseteq$ a set of neighbors in sub-dim
 - Two-level dimensional filtering
 - Find outliers and neighbors early in the sub-dimensional space
- Multiple outlier detection queries scenario
 - Multiple queries – different number of neighbors and distance thresholds
- Streaming scenario
 - Distribution of data points and queries change over time

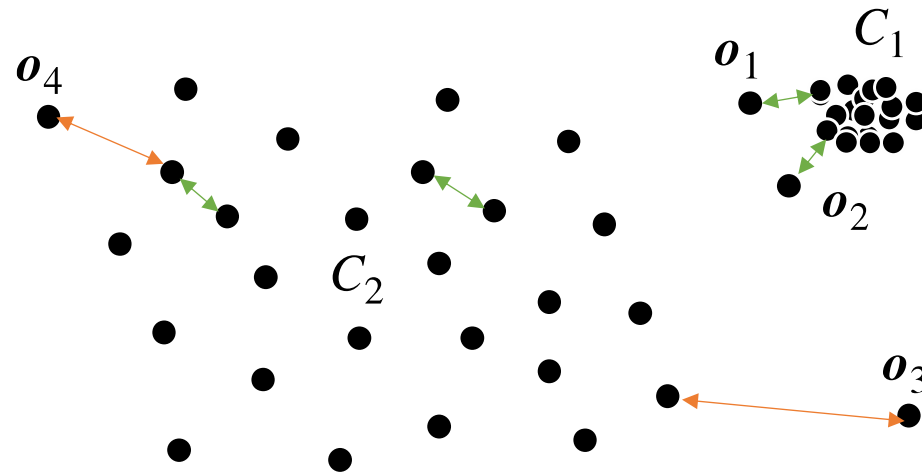
Outline

- ❑ Basic concepts of outlier detection
- ❑ Statistical methods
- ❑ **Proximity-based methods**
 - Distance-based (kNN)
 - **Density-based (LOF, KDE)**
- ❑ Methods for high-dimensional data
- ❑ Outlier detection evaluation

■ Outlier Detection > Proximity > Density-based

□ Local proximity-based outliers

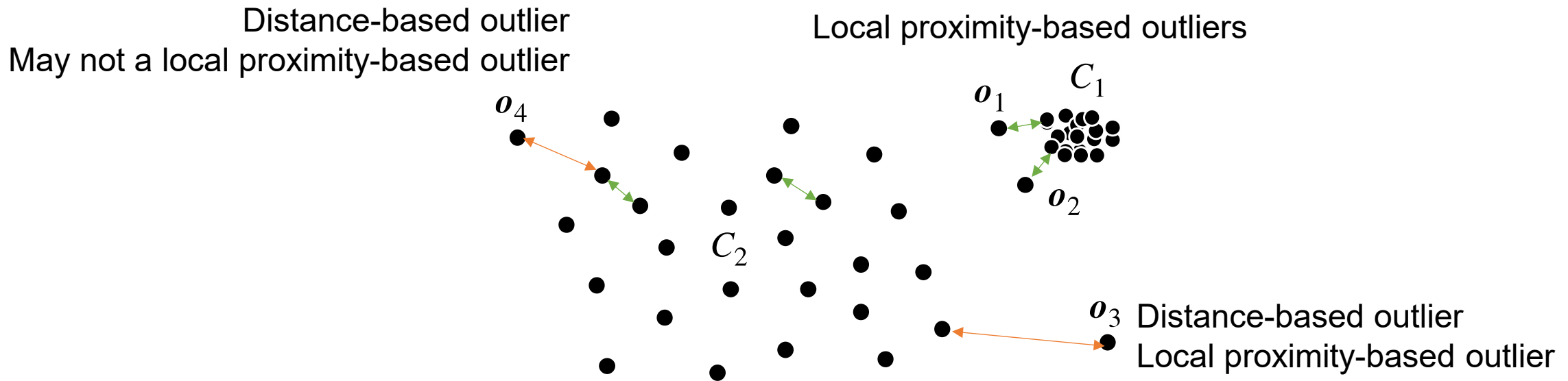
- An abnormal object dissimilar to other objects within its local proximity
- Can be seen as contextual outliers where local proximity defines a context



■ Outlier Detection > Proximity > Density-based

□ Local proximity-based outliers

- An abnormal object dissimilar to other objects within its local proximity
- Can be seen as contextual outliers where local proximity defines a context



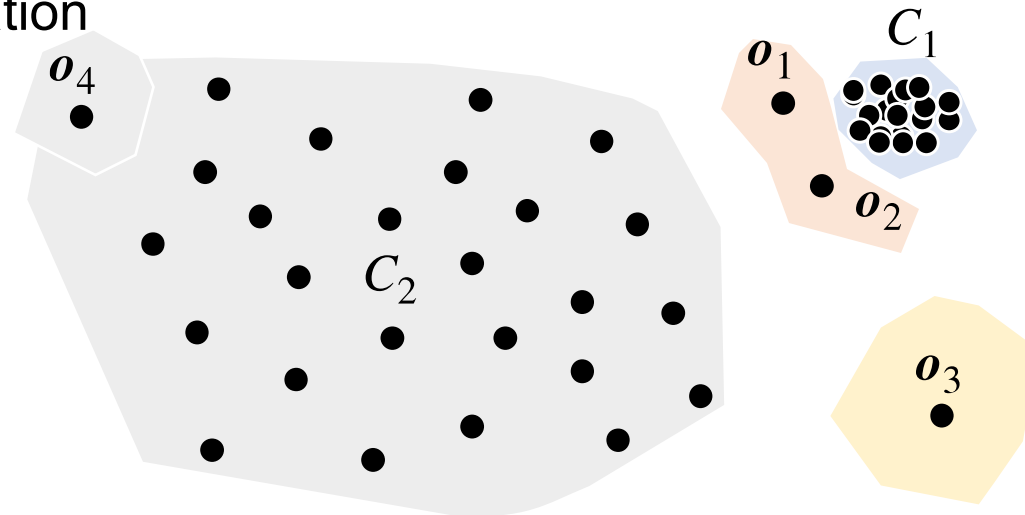
■ Outlier Detection > Proximity > Density-based

□ Local proximity-based outliers

- An abnormal object dissimilar to other objects within its local proximity
- Can be seen as contextual outliers where local proximity defines a context

□ How to measure the local proximity (i.e., relative density) of an object?

- Local outlier factor
- Kernel density estimation

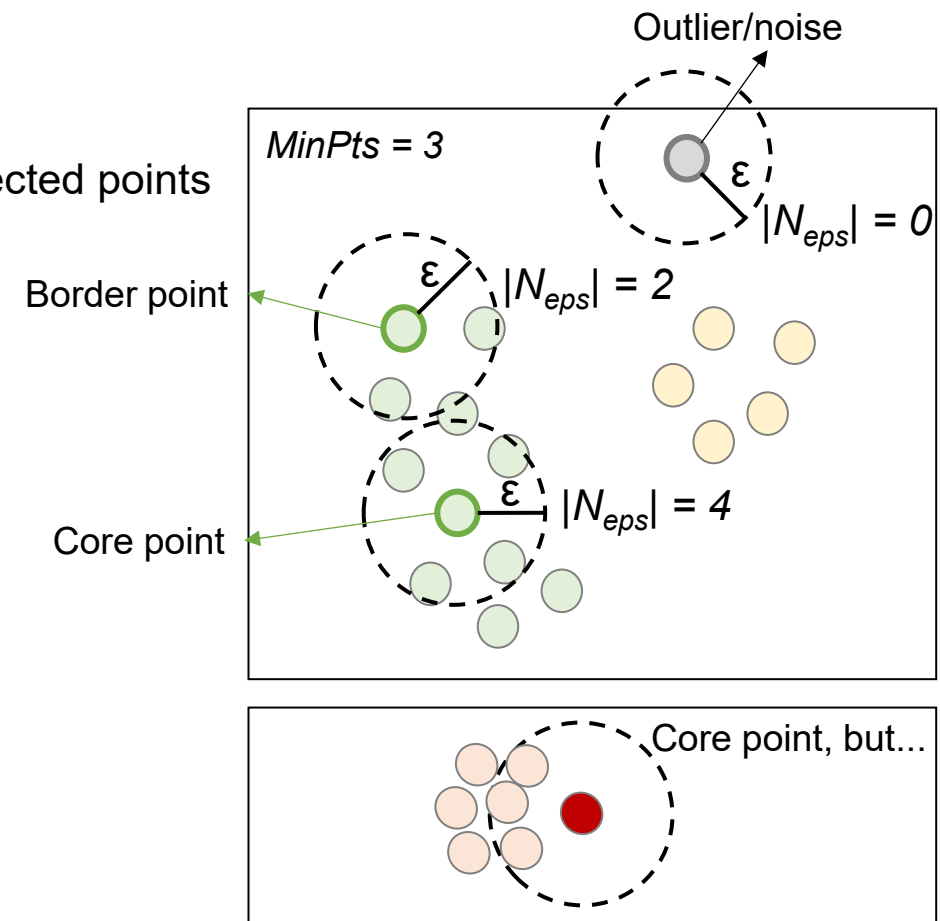


Local proximity as a degree of density

■ Outlier Detection > Proximity > Density-based

□ Local outlier factor (LOF)

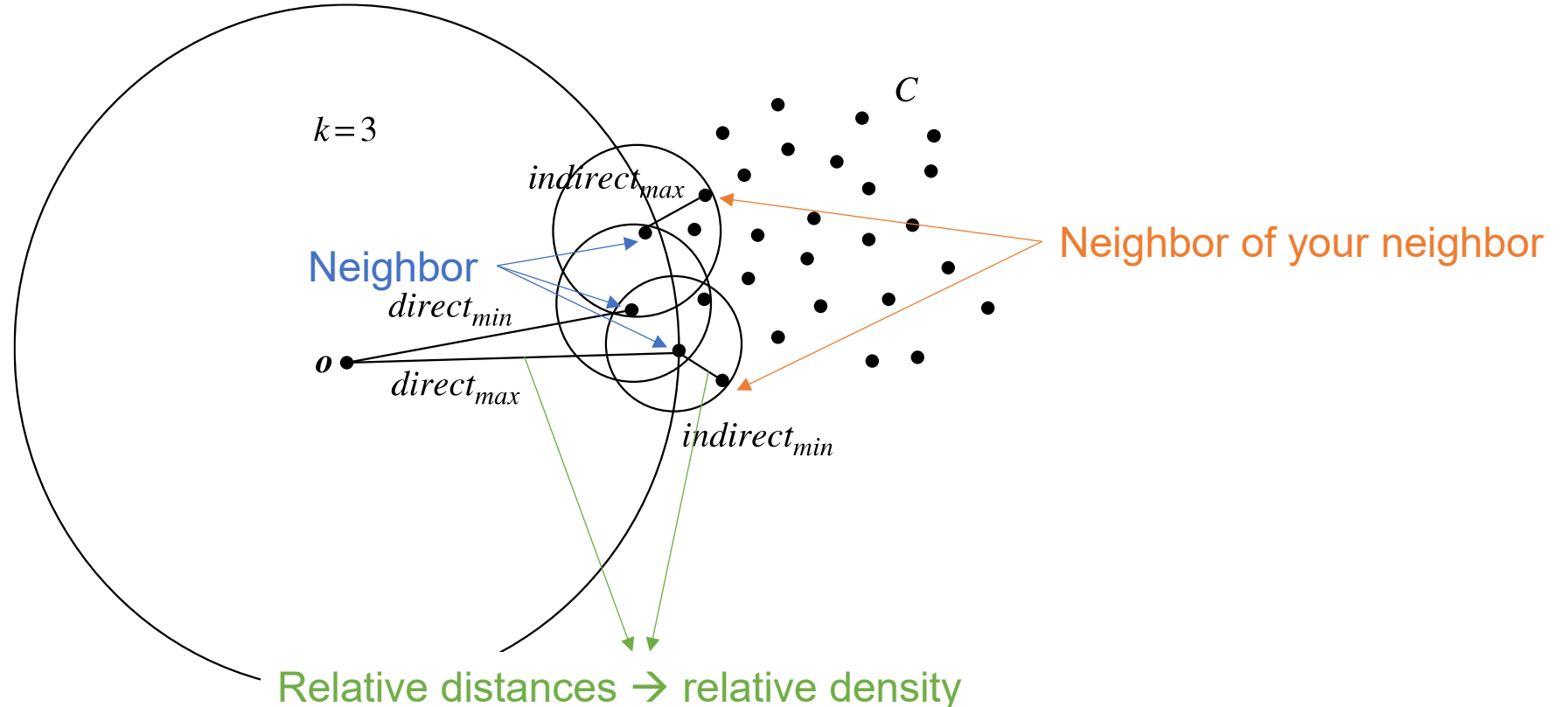
- Define a notion of density with k nearest neighbors (kNN)
- Relevant concept used for clustering?
 - [Recap] DBSCAN
 - A density-based notion of cluster as a maximal set of density-connected points
 - The Eps -neighborhood of a point x : $N_{Eps}(x) = \{p \mid d(p, x) \leq Eps\}$
 - Core point: in a dense neighborhood (i.e., cluster)
i.e., $|N_{Eps}(x)| \geq MinPts$
 - Border point: not a core point, but the neighbor of a core point
i.e., $|N_{Eps}(x)| < MinPts$, but $x \in N_{Eps}(x')$ for a core point x'
 - Outlier/noise: neither a core point nor the neighbor of a core point
 - Outlier/noise is defined by static Eps-neighborhood
- “Relative” density is considered as local proximity



■ Outlier Detection > Proximity > Density-based > LOF

□ Local outlier factor (LOF)

- Define a notion of density with k nearest neighbors (kNN)
- “Relative” density is considered as local proximity



■ Outlier Detection > Proximity > Density-based > LOF

□ Local outlier factor (LOF)

- Define a notion of density with k nearest neighbors (kNN)
- “Relative” density is considered as local proximity

□ Building blocks for local outlier factor (LOF)

1. k-distance
2. k-distance neighborhood
3. Reachability distance
4. Local reachability density
5. Local outlier factor

■ Outlier Detection > Proximity > Density-based > LOF

□ Local outlier factor (LOF)

- Define a notion of density with k nearest neighbors (kNN)
- “Relative” density is considered as local proximity

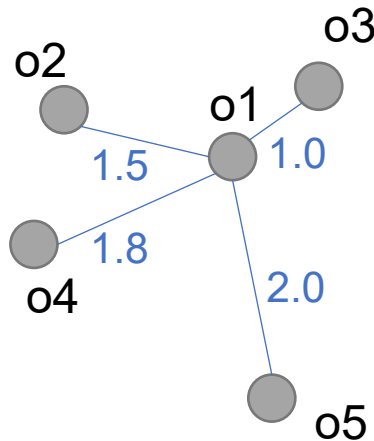
□ Building blocks for local outlier factor (LOF)

1. k -distance $\rightarrow$ *the distance to find your neighbors*
2. k -distance neighborhood $\rightarrow$ *the area your neighbors reside*
3. Reachability distance $\rightarrow$ *relative distance from your neighbors*
4. Local reachability density $\rightarrow$ *local density from relative distances from your neighbors*
5. Local outlier factor $\rightarrow$ *relative density to your neighbors*

■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- **k-distance** → k-distance neighborhood → Reachability distance → Local reachability density → Local outlier factor
- k-distance
 - The k-distance $dist_k(o)$ is the distance between o and its k-nearest neighbor p
 - Formerly, $dist_k(o)$ is the distance $dist(o, p)$ between o and another object $p \in D$ that satisfies
 - There are at least k objects $o' \in D \setminus \{o\}$ such that $dist(o, o') \leq dist(o, p) = dist_k(o)$
 - There are at most $k-1$ objects $o'' \in D \setminus \{o\}$ such that $dist(o, o'') < dist(o, p) = dist_k(o)$



For $k = 3$,

The 3rd nearest neighbor of $o1 \rightarrow p = o4$

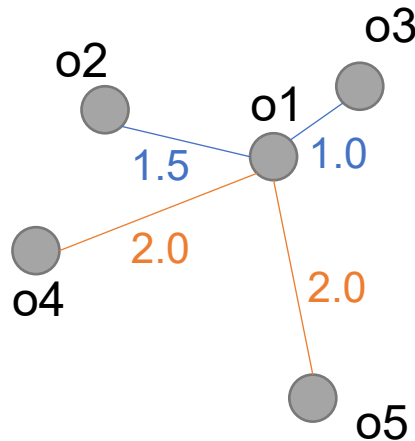
The k-distance $dist_k(o1) = dist(o1, o4) = 1.8$

- There are at least $k=3$ objects ($o2, o3, o4$) ≤ 1.8
- There are at most $k-1=2$ objects ($o2, o3$) < 1.8

■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- **k-distance** → k-distance neighborhood → Reachability distance → Local reachability density → Local outlier factor
- k-distance
 - The k-distance $dist_k(o)$ is the distance between o and its k-nearest neighbor p
 - Formerly, $dist_k(o)$ is the distance $dist(o, p)$ between o and another object $p \in D$ that satisfies
 - There are at least k objects $o' \in D \setminus \{o\}$ such that $dist(o, o') \leq dist(o, p) = dist_k(o)$
 - There are at most $k-1$ objects $o'' \in D \setminus \{o\}$ such that $dist(o, o'') < dist(o, p) = dist_k(o)$



For $k = 3$,

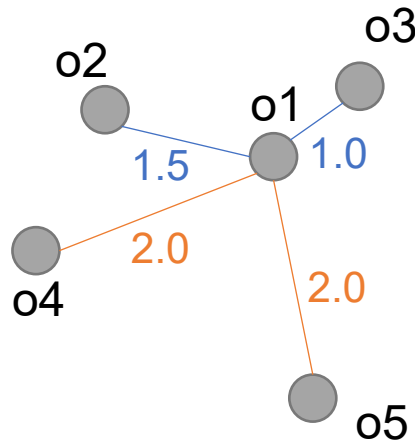
Quiz: What is the k-distance $dist_k(o1)$?

- There are at least k objects $\leq dist_k(o1)$
- There are at most $k-1$ objects $< dist_k(o1)$

■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- **k-distance** → k-distance neighborhood → Reachability distance → Local reachability density → Local outlier factor
- k-distance
 - The k-distance $dist_k(o)$ is the distance between o and its k-nearest neighbor p
 - Formerly, $dist_k(o)$ is the distance $dist(o, p)$ between o and another object $p \in D$ that satisfies
 - There are at least k objects $o' \in D \setminus \{o\}$ such that $dist(o, o') \leq dist(o, p) = dist_k(o)$
 - There are at most $k-1$ objects $o'' \in D \setminus \{o\}$ such that $dist(o, o'') < dist(o, p) = dist_k(o)$
 - Simply speaking, the distance to its k-th nearest neighbor(**s**)



For $k = 3$,

The 3rd nearest neighbors of $o1 \rightarrow p = o4$ and $o5$ (i.e., tie)

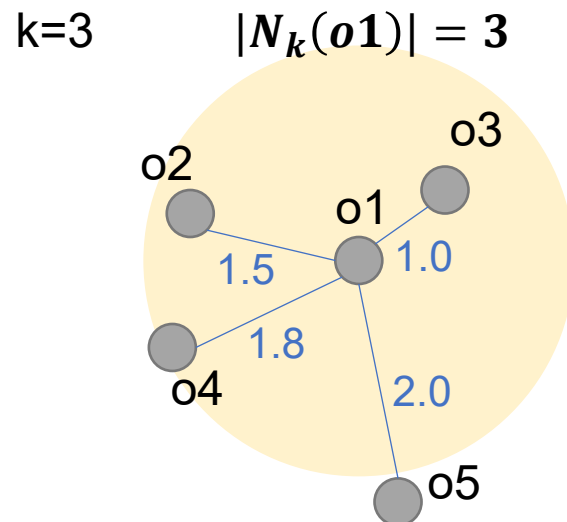
The k-distance $dist_k(o1) = dist(o1, o4) = dist(o1, o5) = 2.0$

■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → **k-distance neighborhood** → Reachability distance → Local reachability density → Local outlier factor
- k-distance neighborhood
 - k-distance neighborhood of o contains all objects of which the distance to o is not greater than $dist_k(o)$

$$N_k(o) = \{o' | o' \in D \setminus \{o\}, dist(o, o') \leq dist_k(o)\}$$



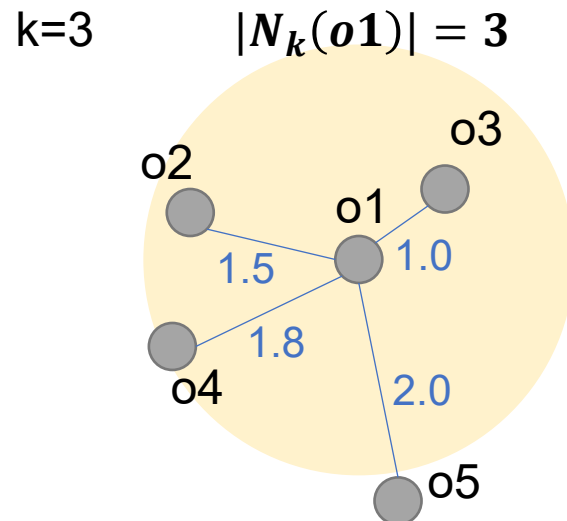
■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → **k-distance neighborhood** → Reachability distance → Local reachability density → Local outlier factor
- k-distance neighborhood
 - k-distance neighborhood of o contains all objects of which the distance to o is not greater than $dist_k(o)$

$$N_k(o) = \{o' | o' \in D \setminus \{o\}, dist(o, o') \leq dist_k(o)\}$$

- **Quiz: Can $N_k(o)$ include more than k objects?**



Outlier Detection > Proximity > Density-based > LOF

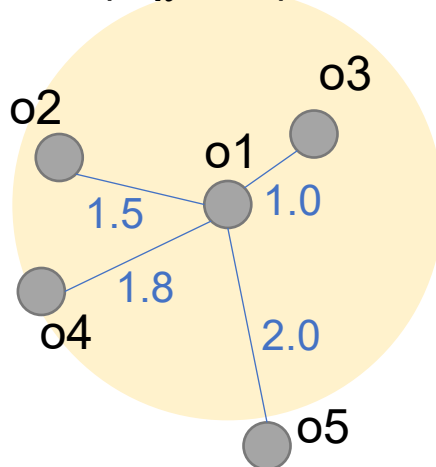
□ Building blocks for local outlier factor (LOF)

- k-distance → **k-distance neighborhood** → Reachability distance → Local reachability density → Local outlier factor
- k-distance neighborhood
 - k-distance neighborhood of o contains all objects of which the distance to o is not greater than $dist_k(o)$

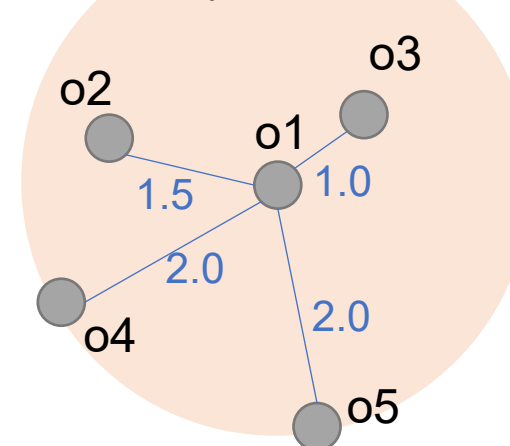
$$N_k(o) = \{o' | o' \in D \setminus \{o\}, dist(o, o') \leq dist_k(o)\}$$

- *Can $N_k(o)$ include more than k objects? → Yes, because $dist_k(o)$ considers the tie*
 - i.e., there are at least k objects $o' \in D \setminus \{o\}$ such that $dist(o, o') \leq dist_k(o)$

k=3 $|N_k(o1)| = 3$



k=3 $|N_k(o1)| = 4$



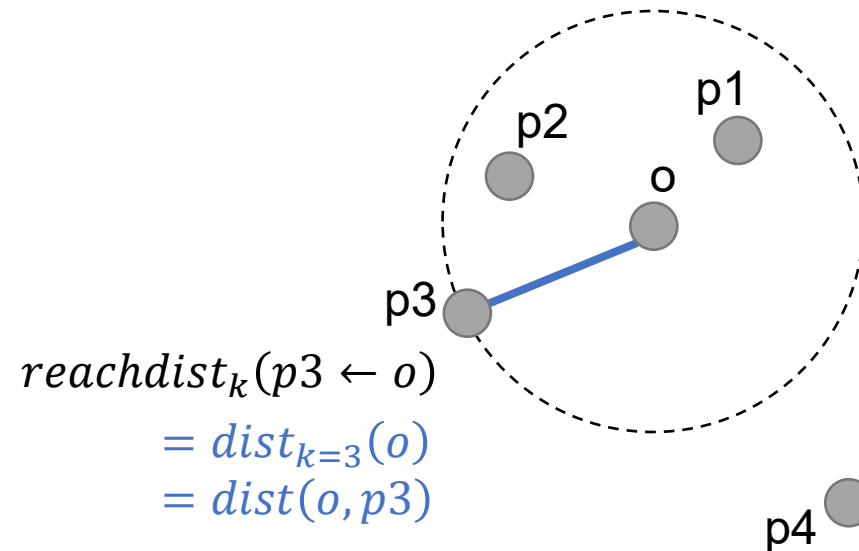
Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → **Reachability distance** → Local reachability density → Local outlier factor
- Reachability distance
 - For o and p, the reachability distance from o to p is $dist_k(o)$ if $dist_k(o) > dist(o, p)$; $dist(o, p)$ otherwise

$$reachdist_k(p \leftarrow o) = \max\{dist_k(o), dist(o, p)\}$$

k=3

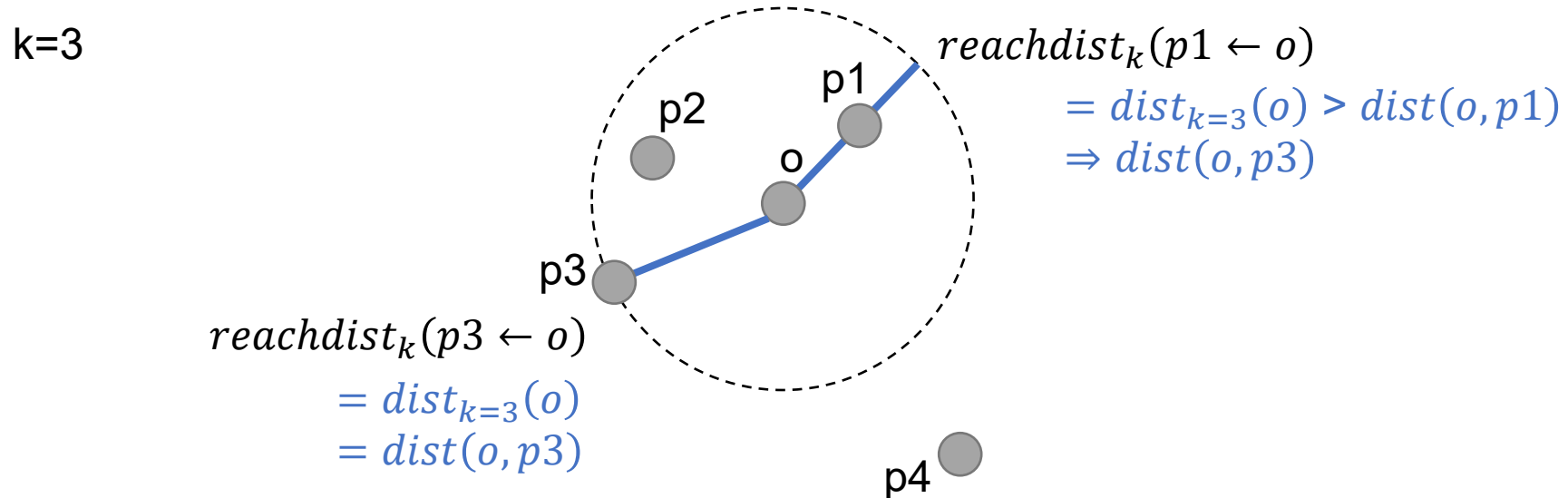


Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → **Reachability distance** → Local reachability density → Local outlier factor
- Reachability distance
 - For o and p, the reachability distance from o to p is $dist_k(o)$ if $dist_k(o) > dist(o, p)$; $dist(o, p)$ otherwise

$$reachdist_k(p \leftarrow o) = \max\{dist_k(o), dist(o, p)\}$$



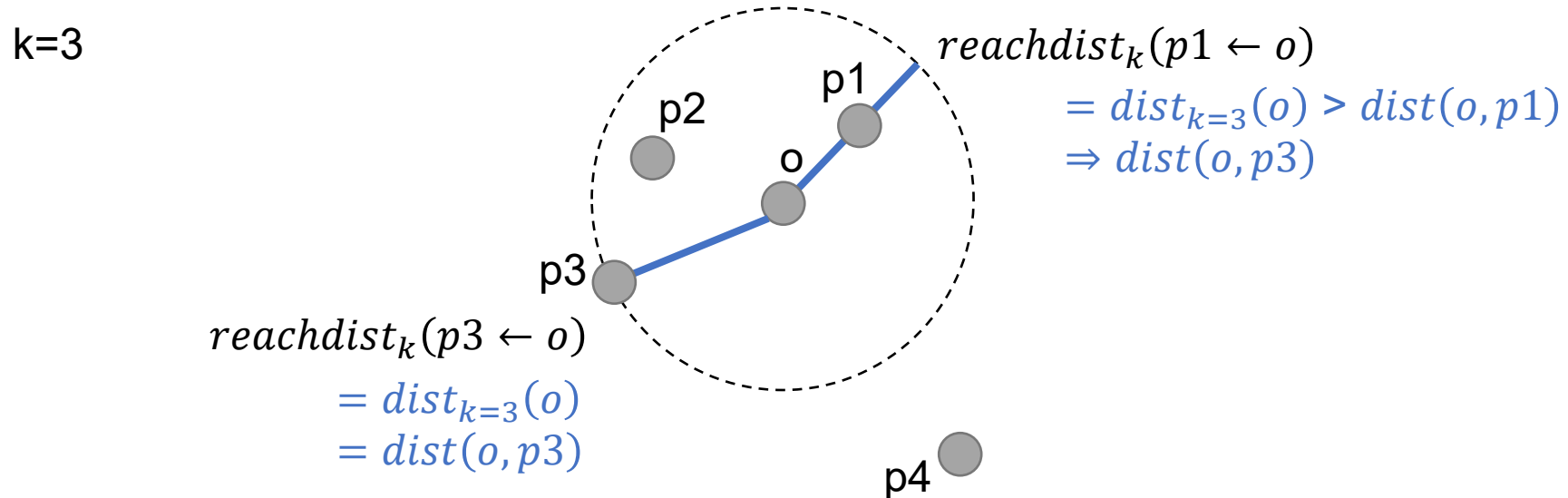
■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → **Reachability distance** → Local reachability density → Local outlier factor
- Reachability distance
 - For o and p, the reachability distance from o to p is $dist_k(o)$ if $dist_k(o) > dist(o, p)$; $dist(o, p)$ otherwise

$$reachdist_k(p \leftarrow o) = \max\{dist_k(o), dist(o, p)\}$$

Quiz: What are $reachdist_k(p2 \leftarrow o)$ and $reachdist_k(p4 \leftarrow o)$?

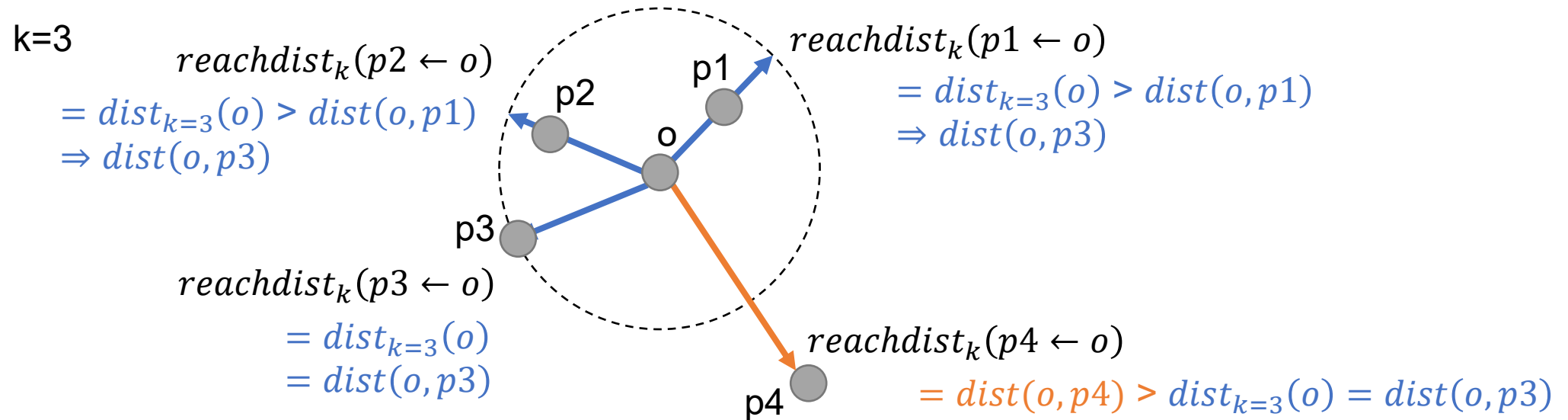


Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → **Reachability distance** → Local reachability density → Local outlier factor
- Reachability distance
 - For o and p, the reachability distance from o to p is $dist_k(o)$ if $dist_k(o) > dist(o, p)$; $dist(o, p)$ otherwise

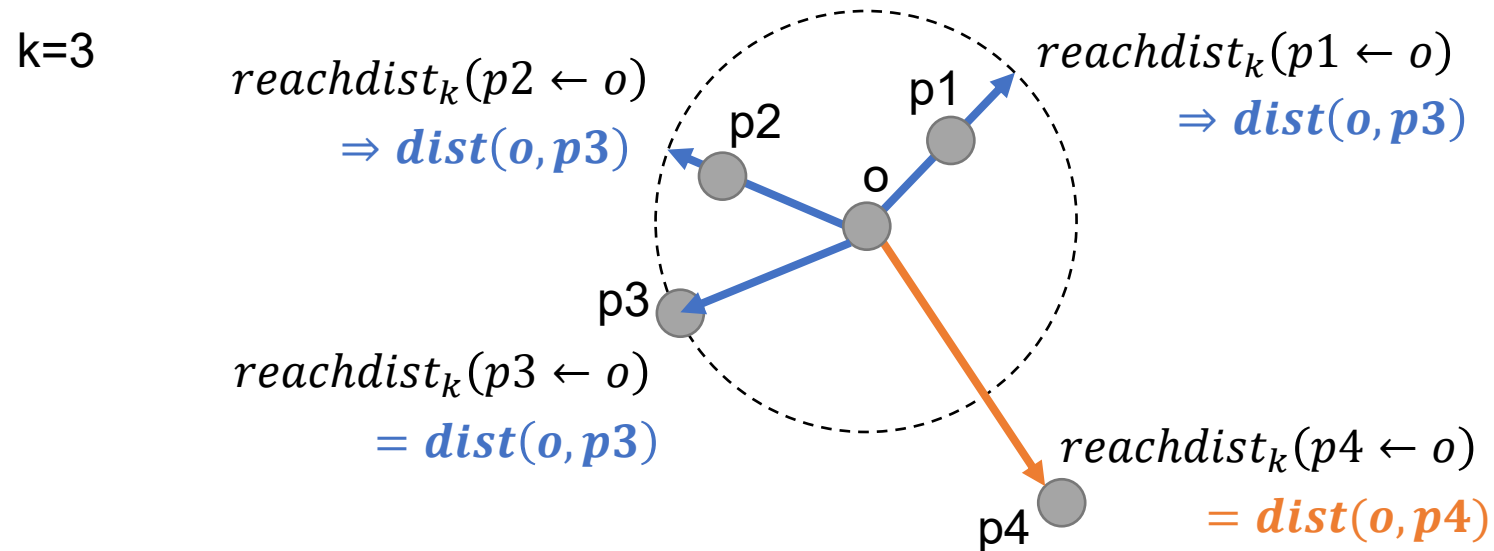
$$reachdist_k(p \leftarrow o) = \max\{dist_k(o), dist(o, p)\}$$



Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

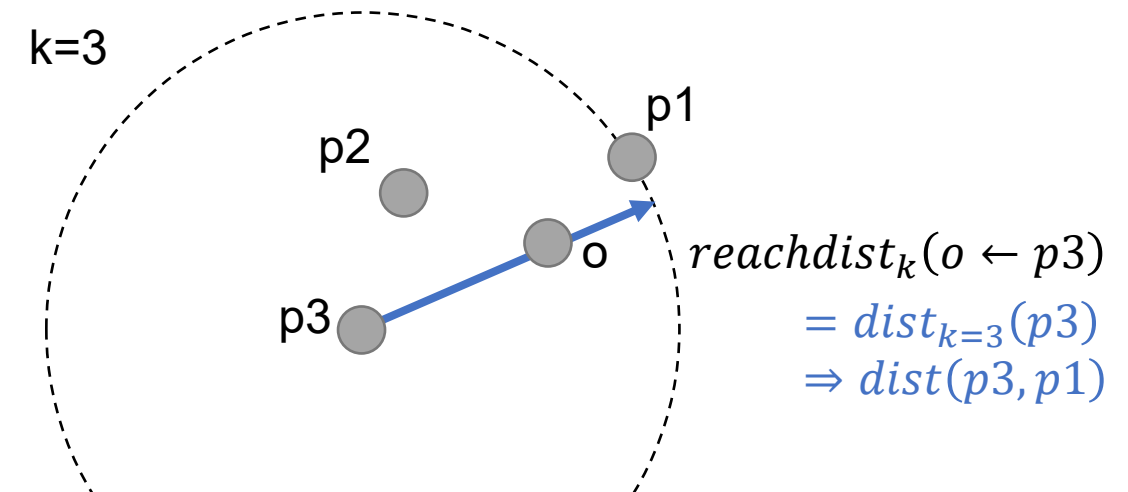
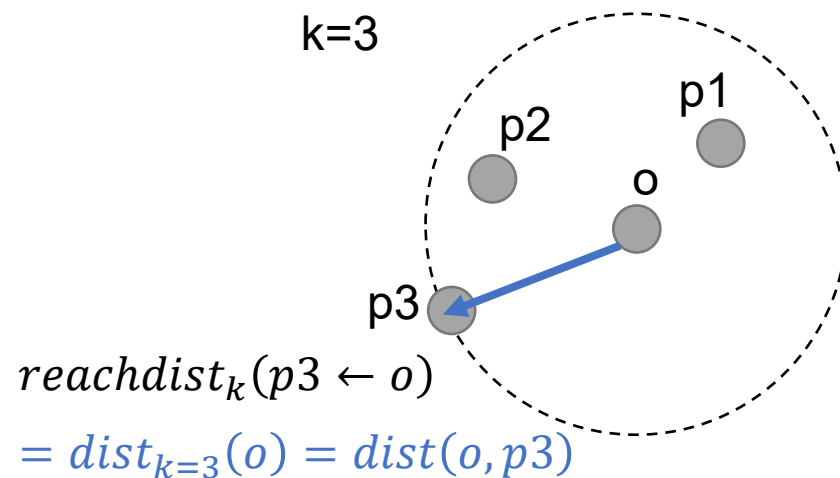
- k-distance → k-distance neighborhood → **Reachability distance** → Local reachability density → Local outlier factor
- Reachability distance
 - For o and p, the reachability distance from o to p is $dist_k(o)$ if $dist_k(o) > dist(o, p)$; $dist(o, p)$ otherwise
- Simply speaking, your neighbors are reachable by $dist_k(o)$ but not others outside the neighborhood



■ Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → **Reachability distance** → Local reachability density → Local outlier factor
 - Reachability distance
 - For o and p, the reachability distance from o to p is $dist_k(o)$ if $dist_k(o) > dist(o, p)$; $dist(o, p)$ otherwise
- $$reachdist_k(p \leftarrow o) = \max\{dist_k(o), dist(o, p)\}$$
- Simply speaking, your neighbors are reachable by $dist_k(o)$ but not others outside the neighborhood
 - Reachability distance is not symmetric, that is, in general, $reachdist_k(p \leftarrow o) \neq reachdist_k(o \leftarrow p)$

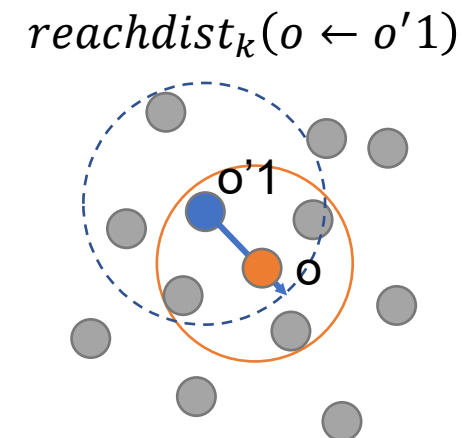
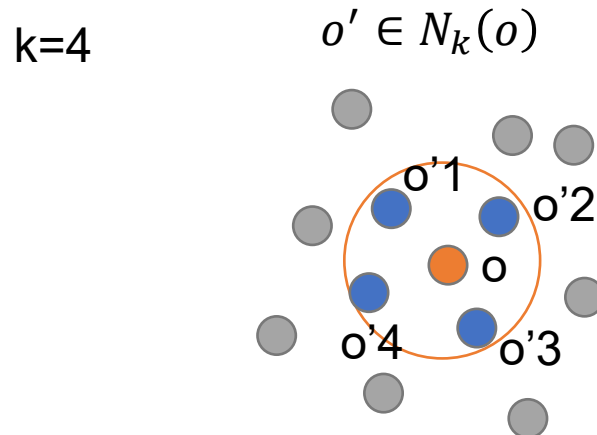


Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → Reachability distance → **Local reachability density** → Local outlier factor
- Local reachability density of an object o
 - The inverse of the average reachability distance of the neighbors in k-distance neighborhood of o

$$lrd_k(o) = \frac{|N_k(o)|}{\sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o')} = \left(\frac{1}{|N_k(o)|} \sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o') \right)^{-1}$$



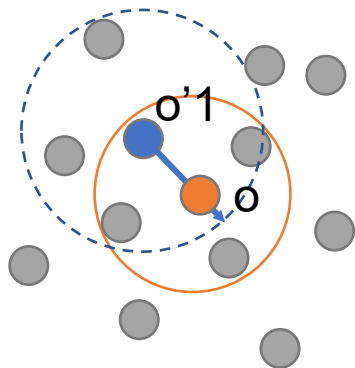
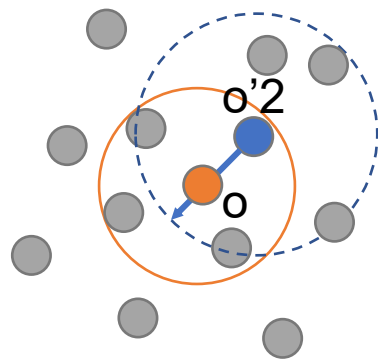
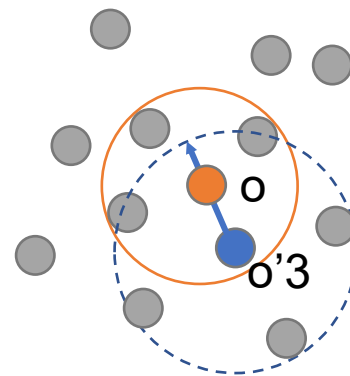
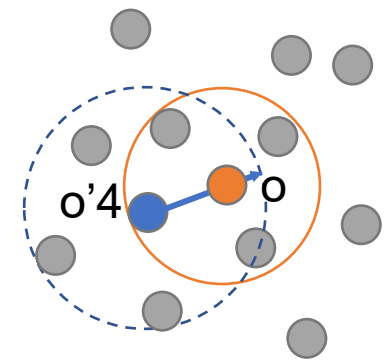
Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

- k-distance → k-distance neighborhood → Reachability distance → **Local reachability density** → Local outlier factor
- Local reachability density of an object o
 - The inverse of the average reachability distance of the neighbors in k-distance neighborhood of o

$$lrd_k(o) = \frac{|N_k(o)|}{\sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o')} = \left(\frac{1}{|N_k(o)|} \sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o') \right)^{-1}$$

k=4

 $reachdist_k(o \leftarrow o'1)$  $reachdist_k(o \leftarrow o'2)$  $reachdist_k(o \leftarrow o'3)$  $reachdist_k(o \leftarrow o'4)$ 

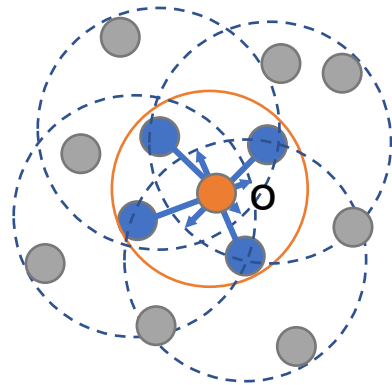
Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

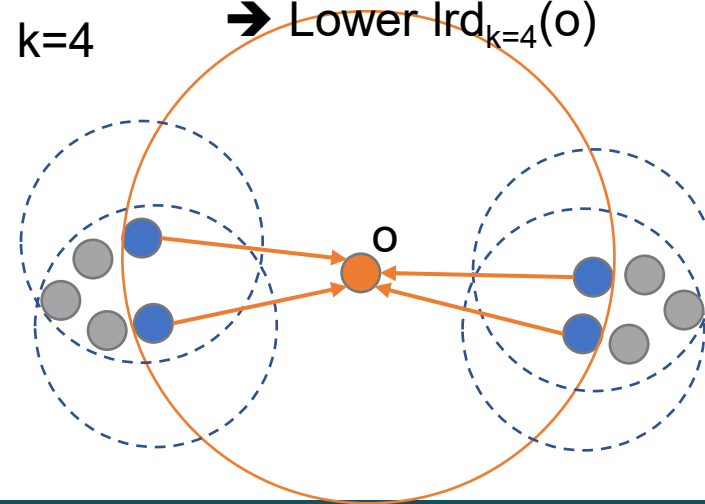
- k-distance → k-distance neighborhood → Reachability distance → **Local reachability density** → Local outlier factor
- Local reachability density of an object o
 - The inverse of the average reachability distance of the neighbors in k-distance neighborhood of o

$$lrd_k(o) = \frac{|N_k(o)|}{\sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o')} = \left(\frac{1}{|N_k(o)|} \sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o') \right)^{-1}$$

Smaller $reachdist_{k=4}(o \leftarrow o')$
 $k=4$ → Higher $lrd_{k=4}(o)$



Larger $reachdist_{k=4}(o \leftarrow o')$
 $k=4$ → Lower $lrd_{k=4}(o)$



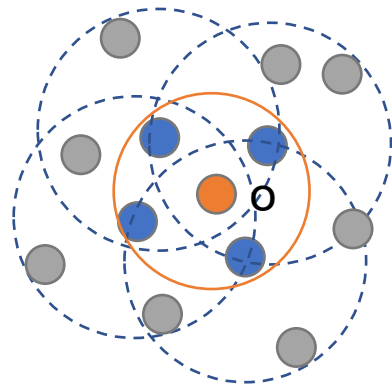
Outlier Detection > Proximity > Density-based > LOF

□ Building blocks for local outlier factor (LOF)

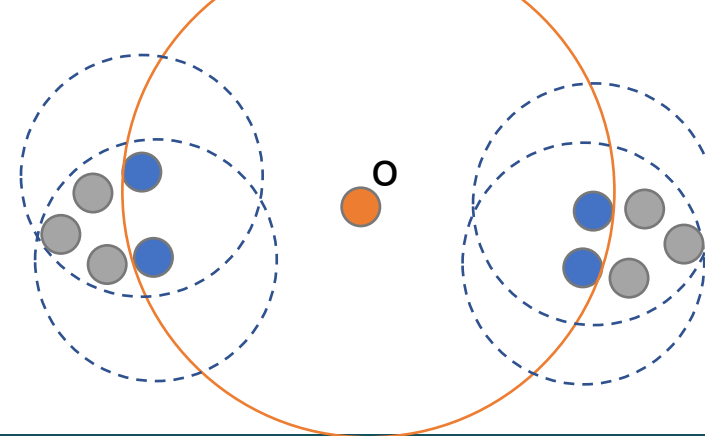
- k-distance → k-distance neighborhood → Reachability distance → Local reachability density → **Local outlier factor**
- Local outlier factor of an object o
 - The average of the ratio of the local reachability density of o and those of the neighbors in k-distance neighborhood of o

$$LOF_k(o) = \frac{1}{|N_{k(o)}|} \sum_{o' \in N_{k(o)}} \frac{lrd_k(o')}{lrd_k(o)} \left(= \frac{\frac{1}{|N_{k(o)}|} \sum_{o' \in N_{k(o)}} lrd_k(o')}{lrd_k(o)} \right)$$

k=4
 $lrd_{k=4}(o) \approx lrd_{k=4}(o')$
 → Lower LOF(o)



k=4
 $lrd_{k=4}(o) \ll lrd_{k=4}(o')$
 → Higher LOF(o)



■ Outlier Detection > Proximity > Density-based > LOF

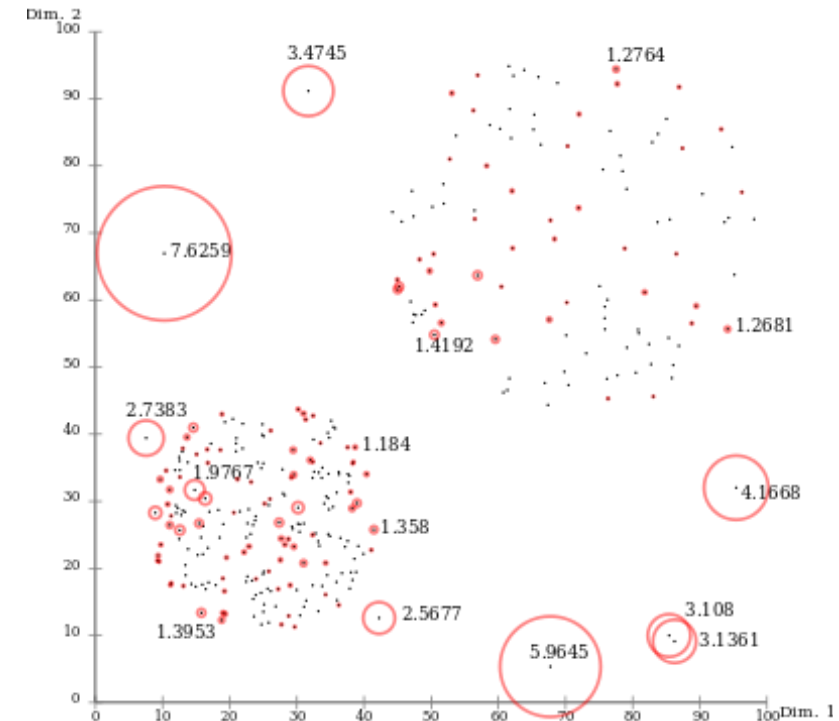
□ Local outlier factor (LOF) of an object o

- The average of the ratio of the local reachability density of o and those of k-distance neighborhood of o

$$LOF_k(o) = \frac{1}{|N_k(o)|} \sum_{o' \in N_k(o)} \frac{lrd_k(o')}{lrd_k(o)}$$

□ Interpretation of LOF

- $LOF(o) < 1$: an object o is in a relatively denser region
- $LOF(o) > 1$: an object o is in a relatively sparser region

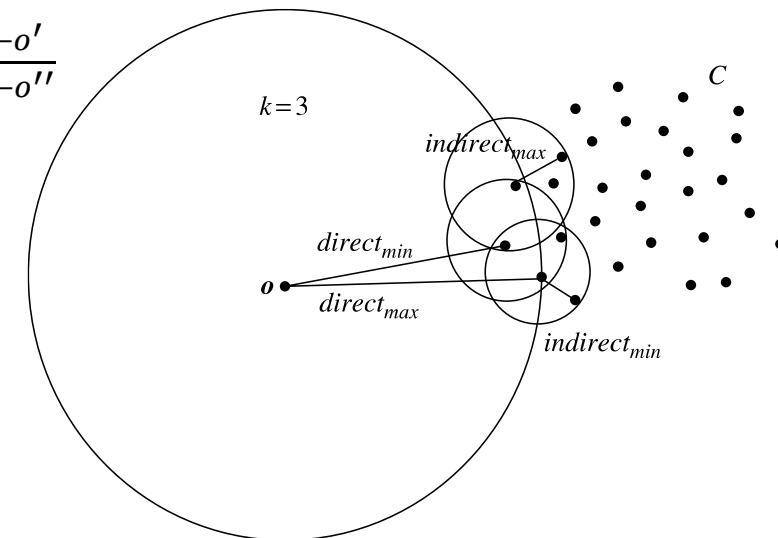
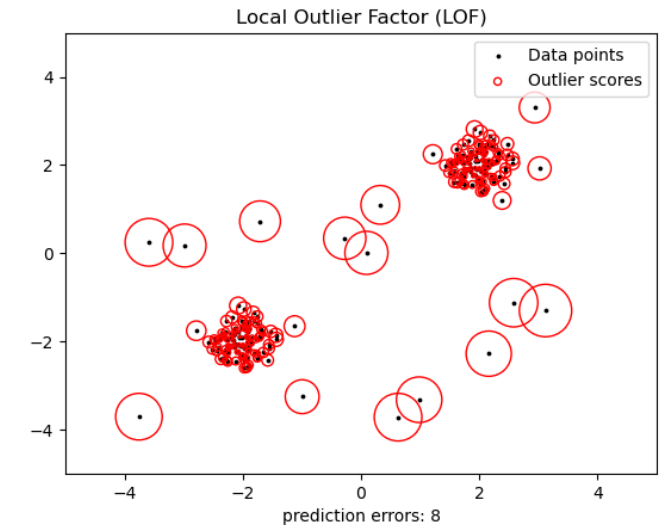


Outlier Detection > Proximity > Density-based > LOF

□ Properties of LOF

- For an object deep within a consistent cluster, LOF is close to 1
 - The density of the object is comparable to those of its neighbors
 - The objects inside clusters, whether dense or sparse, can not be outliers
- Bounds on LOF with the reachability distance to (indirect) neighbors

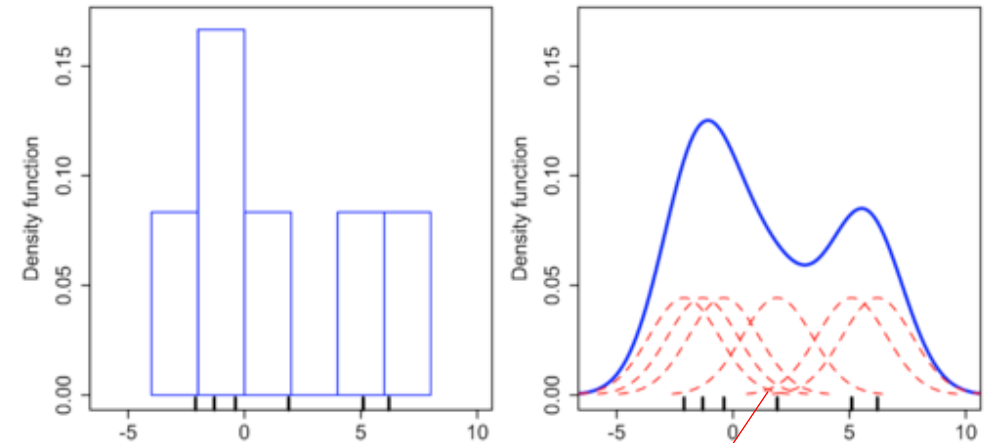
- $LOF_k(o) = \frac{1}{|N_k(o)|} \sum_{o' \in N_k(o)} \frac{lrd_k(o')}{lrd_k(o)}$
 - Roughly, $LOF_k(o) \rightarrow \frac{\text{local density of } o'}{\text{local density of } o} \rightarrow \frac{(\text{local distance } o' \leftarrow o'')^{-1}}{(\text{local distance } o \leftarrow o')^{-1}} \rightarrow \frac{\text{local distance } o \leftarrow o'}{\text{local distance } o' \leftarrow o''}$
- LOF(o) is bounded by $\frac{direct_{min}(o)}{indirect_{max}(o)} \leq LOF(o) \leq \frac{direct_{max}(o)}{indirect_{min}(o)}$
 - $direct_{min}(o) = \min\{reachdist_k(o \leftarrow o') | o' \in N_k(o)\}$
 - $direct_{max}(o) = \max\{reachdist_k(o \leftarrow o') | o' \in N_k(o)\}$
 - $indirect_{min}(o) = \min\{reachdist_k(o' \leftarrow o'') | o' \in N_k(o) \text{ and } o'' \in N_k(o')\}$
 - $indirect_{max}(o) = \max\{reachdist_k(o' \leftarrow o'') | o' \in N_k(o) \text{ and } o'' \in N_k(o')\}$



■ Outlier Detection > Proximity > Density-based > KDE

□ [Recap] Kernel density estimation (KDE)

- A non-parametric method to estimate the probability density function (PDF) of a dataset X
- Given a point x_i , the kernel density estimator of X computes how likely x_i is drawn from X
 - The density at x_i in X is $\tilde{f}(x_i) = \frac{1}{m} \sum_{j=1..m} K_h(|x_i - kc_j|)$
- Kernel center kc
 - Each kernel center kc_j carries a kernel function K_h .
 - $\text{dist}(kc_j, x_i)$ gives the density contribution to x_i by kc_j
 - Typically all data points
- Kernel function $K_h(u)$
 - A non-negative real-valued integrable function satisfying
 - $\int_{-\infty}^{\infty} K(u) du = 1$ and $K(-u) = K(u)$ for any u
 - Gaussian kernel function is the most popular one
 - h is the bandwidth serving as a smoothing parameter



$$K_{\text{gauss},h}(u) := \frac{1}{(2\pi)^{d/2} h^d} e^{-\frac{1}{2} \frac{u^2}{h^2}}$$

■ Outlier Detection > Proximity > Density-based > KDE

□ KDE-based local outlier score

- Replaces LOF's density estimate by a variable-width Gaussian kernel density estimation (KDE)

$$\bullet \text{ } lrd_k(o) = \frac{|N_k(o)|}{\sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o')} = \left(\frac{1}{|N_k(o)|} \sum_{o' \in N_k(o)} reachdist_k(o \leftarrow o') \right)^{-1}$$

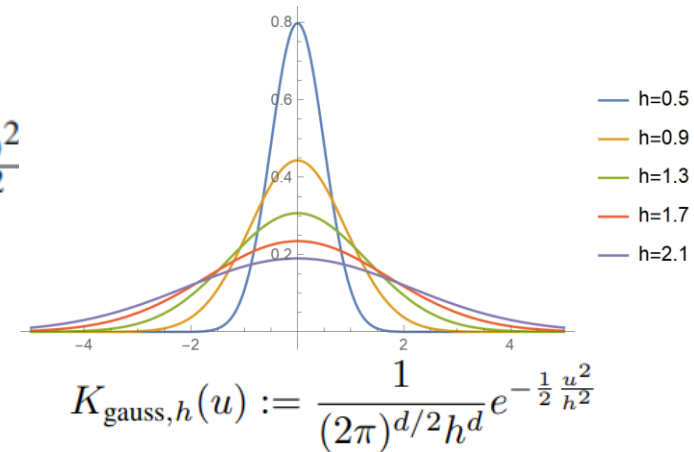
- Local density estimation (LDE)

$$LDE_d(p) := \frac{1}{|kNN(p)|} \sum_{o \in kNN(p)} \frac{1}{(2\pi)^{d/2} (h \cdot k\text{-dist}(p))^d} e^{-\frac{1}{2} \frac{reach\text{-}dist(p,o)^2}{(h \cdot k\text{-dist}(p))^2}}$$

- Local density factor (LDF)

- c is a scaling factor to handle very small or zero LDE(p)

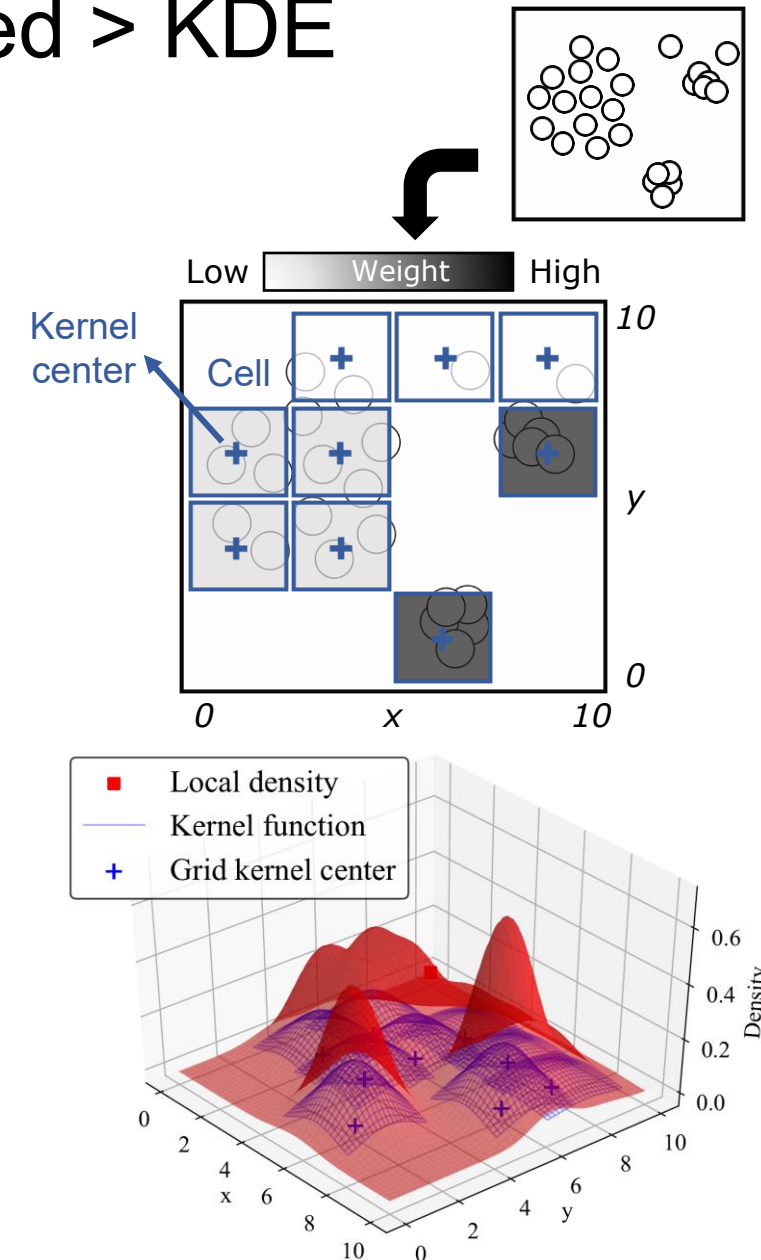
$$LDF(p) = \frac{\frac{1}{k} \sum_{o \in kNN(p)} LDE(o)}{LDE(p) + \frac{c}{k} \sum_{o \in kNN(p)} LDE(o)}$$



■ Outlier Detection > Proximity > Density-based > KDE

□ Approximate KDE-based local outlier score

- Considering all objects as kernel centers is expensive
 - $O(N^2)$ for density estimation
- Approximate kernel center AKC
 - Abstract kernel center (e.g., micro-cluster or cell centroids)
 - Sampled kernel center (e.g., randomly sampled from data points)
- Local density estimation
 - Estimates the density of x_i with only the local AKCs
 - $\tilde{f}(x_i) = \frac{1}{k} \sum_{j=1..k} w_j K_h(d(x_i - kc_j|)), kc_j \in \text{kNN}_{\text{AKC}}(x_i)$
- Local outlier score
 - Relative score: $\frac{1}{k} \sum_{kc_j \in \text{kNN}_{\text{AKC}}(x_i)} \frac{\tilde{f}(kc_j)}{\tilde{f}(x_i)}$
 - Z-score: $z - \text{score}(\tilde{f}(x_i), \{\tilde{f}(kc_j) \mid \forall kc_j \in \text{kNN}_{\text{AKC}}(x_i)\})$



Outline

- ❑ Basic concepts of outlier detection
- ❑ Statistical methods
- ❑ Proximity-based methods
- ❑ **Methods for high-dimensional data**
 - Proximity-based
 - Angle-based
 - Ensemble-based
- ❑ Outlier detection evaluation

■ Outlier Detection > High-dimensional Data

□ Challenges

- Data subspace
 - Since many features (dimensions) are involved, proper interpretation of the outliers is more useful
 - Identifying specific subspaces that manifest the outliers helps users understand the meaning of outliers
 - Using a fixed-distance threshold against all subspaces may not be a good idea
- Data sparsity
 - The distance between objects becomes heavily dominated by noise as the dimensionality increases
- Scalability to dimensionality
 - Exhaustive combinatorial exploration of the search space is not scalable

■ Outlier Detection > High-dimensional Data

□ Approaches

- Proximity-based
- Angle-based
- Ensemble-based

■ Outlier Detection > High-dim > Proximity-based

□ Extend the conventional proximity-based methods

- Try to overcome the curse of dimensionality for proximity measures in high-dimensional spaces
- Use alternative measures or construct subspaces and detect outliers there

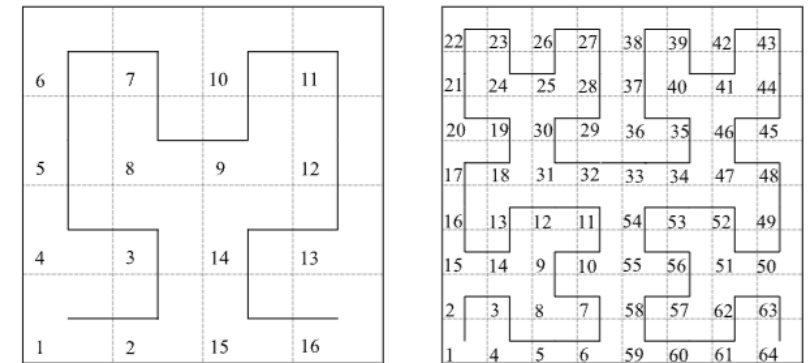
□ Dimensionality reduction

- Principal component analysis (PCA), matrix factorization, ...

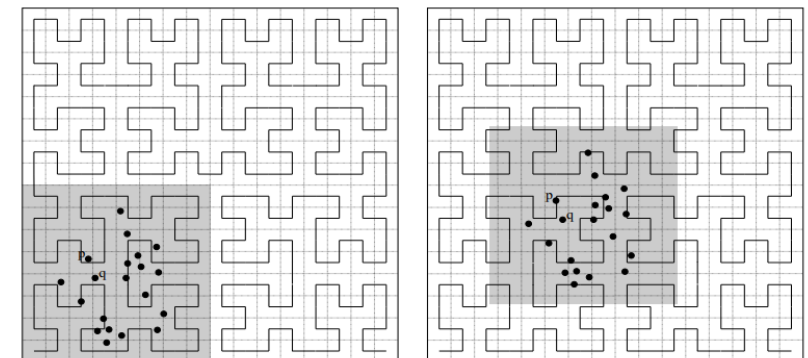
□ Space-filling curve (serialization)

- HilOut finds distance-based outliers, but uses the ranks of distance instead of the absolute value
 - Find the approximate k-nearest neighbors of o , denoted by $nn_1(o), \dots, nn_k(o)$
 - Space-filling curve is used to get one-dimensional distances
 - The outlier score of object o is $w(o) = \sum_{i=1}^k \text{dist}(o, nn_i(o))$
 - The top-N scored objects are output as outliers

Hilbert curve (2D → 1D)



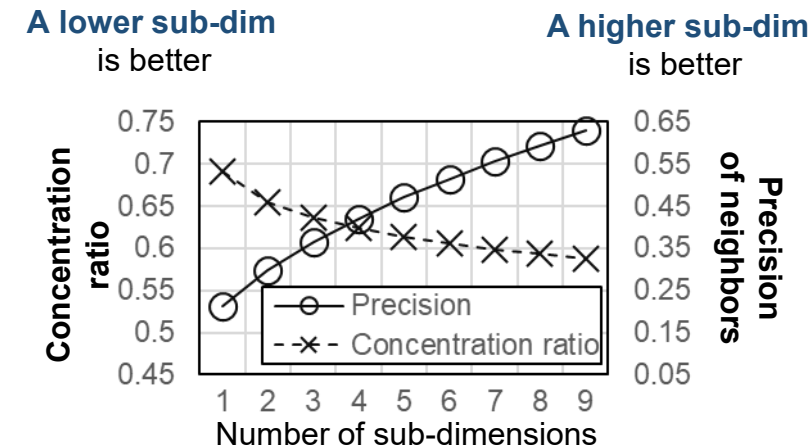
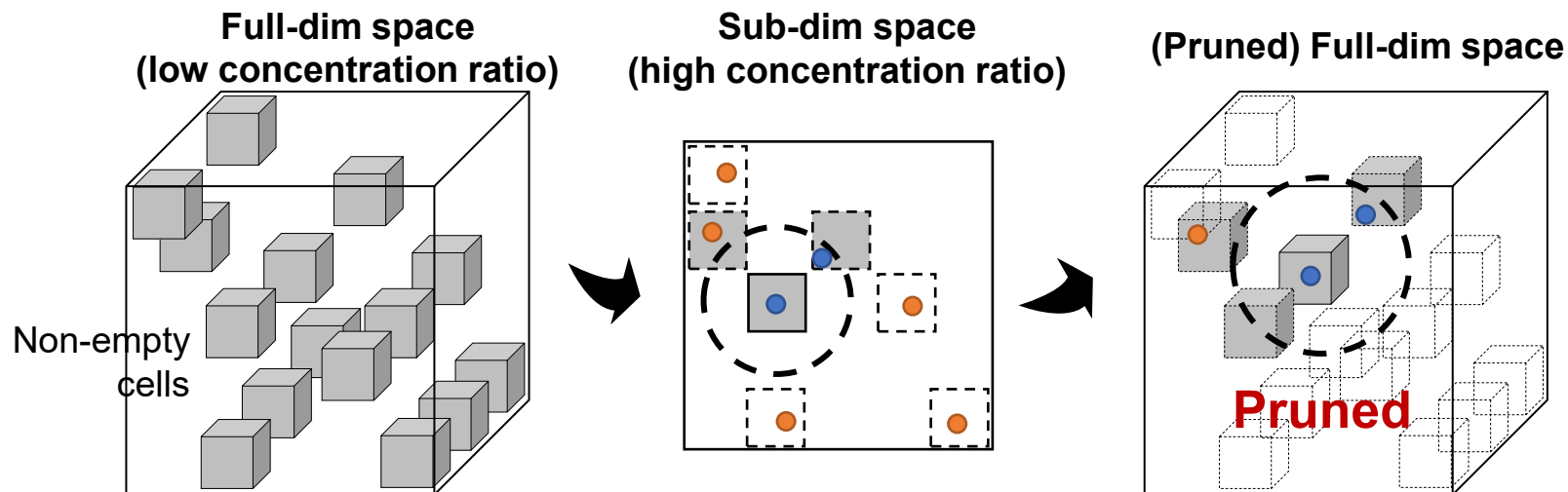
Shifts of the space-filling curves



Outlier Detection > High-dim > Proximity-based

□ Subspace-based pruning

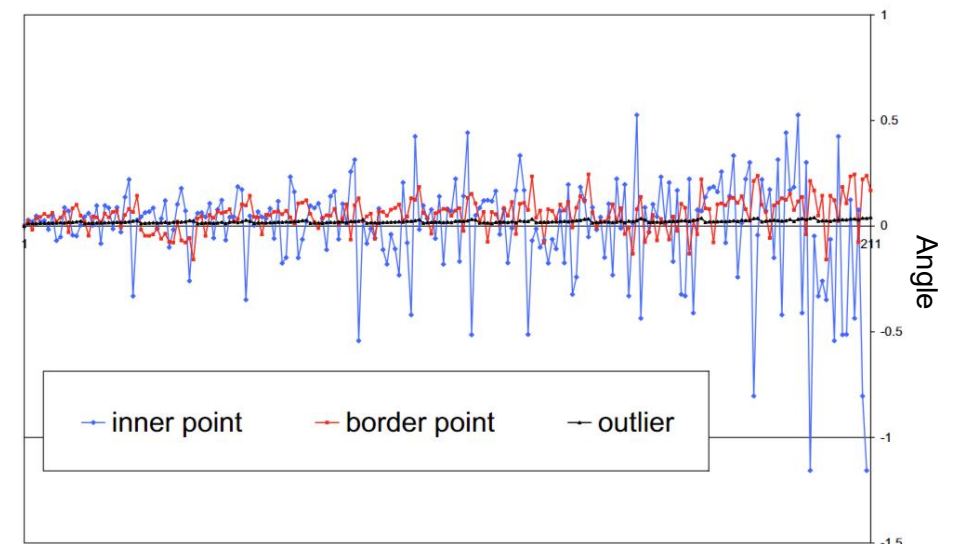
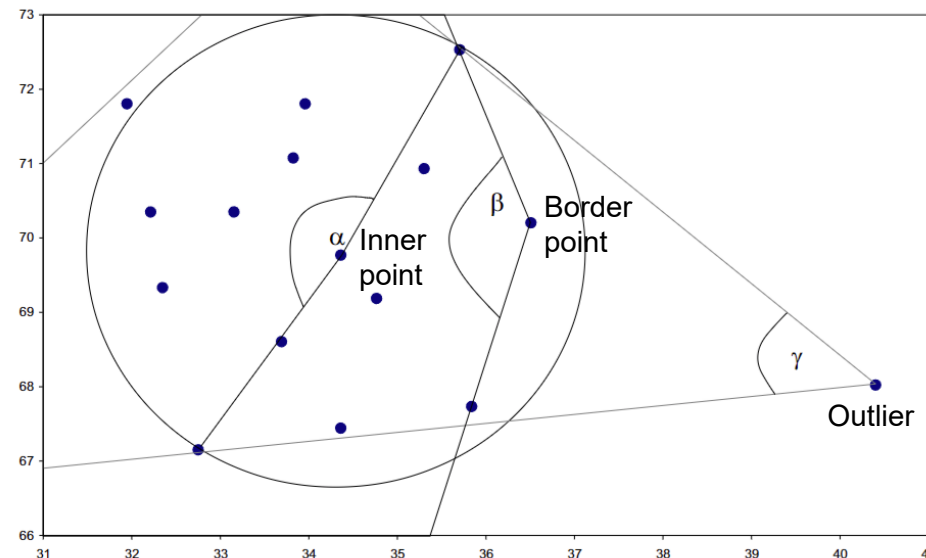
- A distance-based outlier o defined by K neighbors in distance of R . i.e., $|\{o' | \text{dist}(o, o') \leq R\}| \leq K$
 - Assume that Euclidean distance is used
- Downward-closure property of neighbors (feat. downward-closure property of frequent itemsets)
 - Distance in sub-dim $\leq$ distance in full-dim $\Rightarrow$ A set of neighbors in full-dim $\subseteq$ a set of neighbors in sub-dim
- Find outlier-guaranteed cells and/or neighbor candidates early in the sub-dimensional space
 - The optimal number of sub-dimensions depend on the concentration of data over dimensions



■ Outlier Detection > High-dim > Angle-based

□ Angle-based outlier detection (ABOD)

- A direction-focused measure that does not degrade in high-dimensional data
- Idea
 - An object is abnormal if most other objects are located in similar directions
 - An object is normal if many other objects are located in varying directions



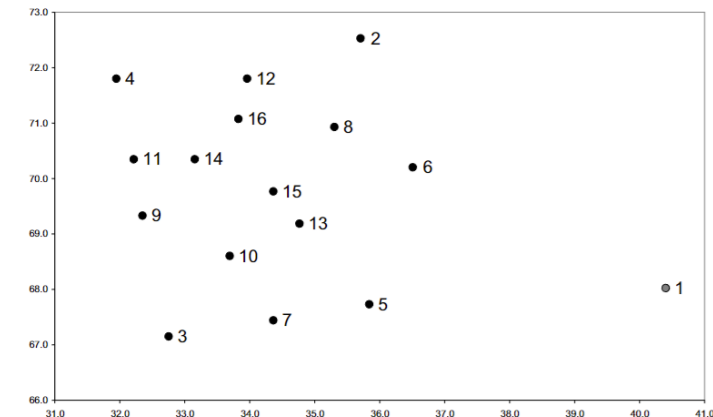
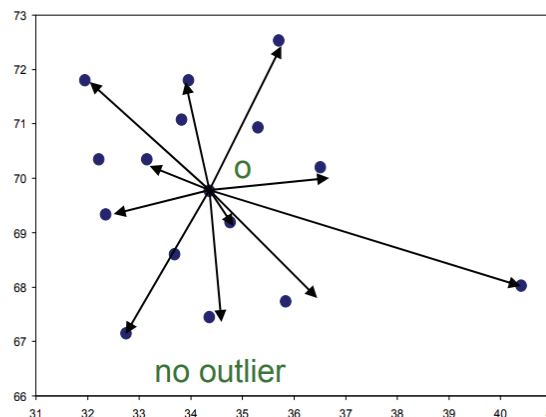
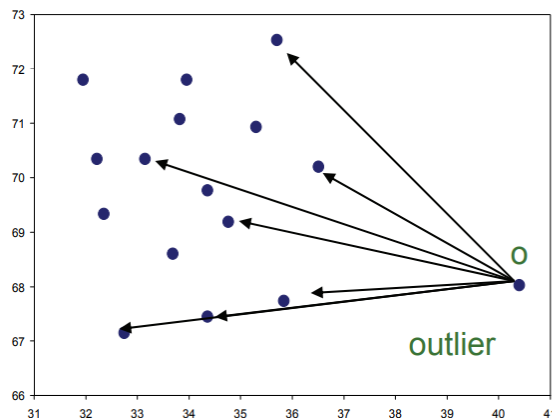
Outlier Detection > High-dim > Angle-based

□ Angle-based outlier detection (ABOD)

- A direction-focused measure that does not deteriorate in high-dimensional data
- Given a set of objects D , for an object $o \in D$, the angle-based outlier factor (ABOF):

$$ABOF(o) = VAR_{x,y \in D \setminus \{o\}} \left(\frac{\cos\theta(\overrightarrow{ox}, \overrightarrow{oy})}{d(o,x)d(o,y)} \right) = VAR_{x,y \in D \setminus \{o\}} \left(\frac{\overrightarrow{ox} \cdot \overrightarrow{oy}}{d(o,x)d(o,y)} \times \frac{1}{d(o,x)d(o,y)} \right) = VAR_{x,y \in D \setminus \{o\}} \left(\frac{\langle \overrightarrow{ox}, \overrightarrow{oy} \rangle}{d(o,x)^2 d(o,y)^2} \right),$$

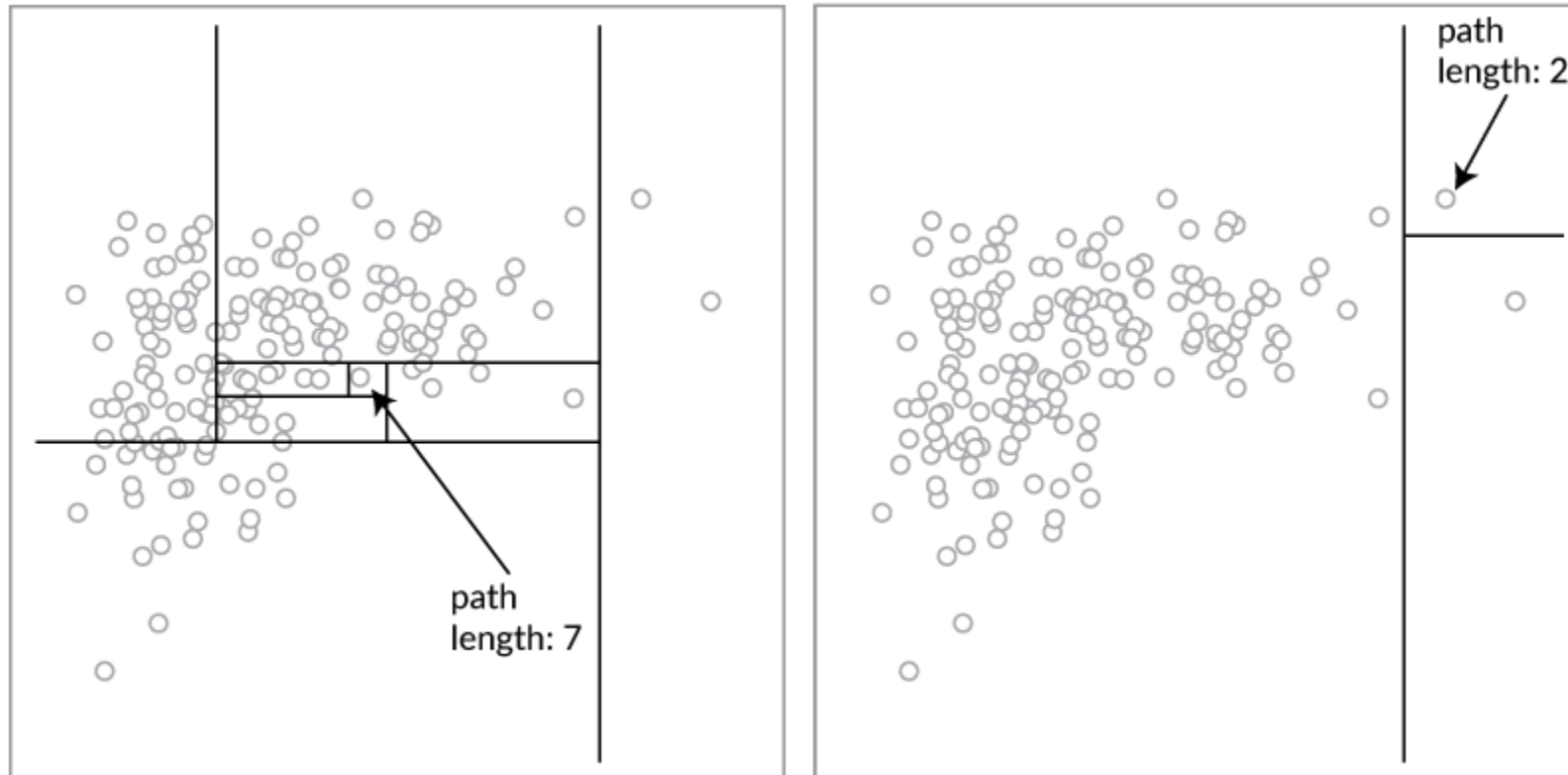
- The angle between $\overrightarrow{ox}$, $\overrightarrow{oy}$ is further weighted by their distances
- Objects with smaller ABOF values become outliers
 - The farther away an object is from others and the smaller the variance of the angles of the object



■ Outlier Detection > High-dim > Ensemble-based

□ Isolation forest

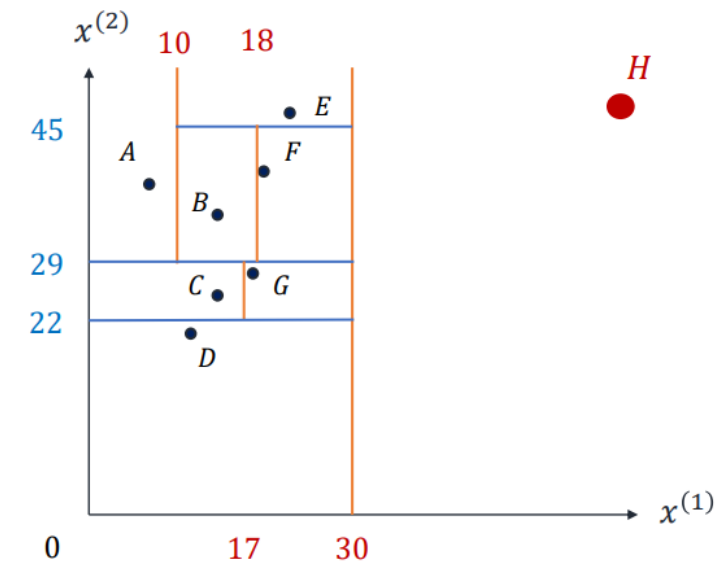
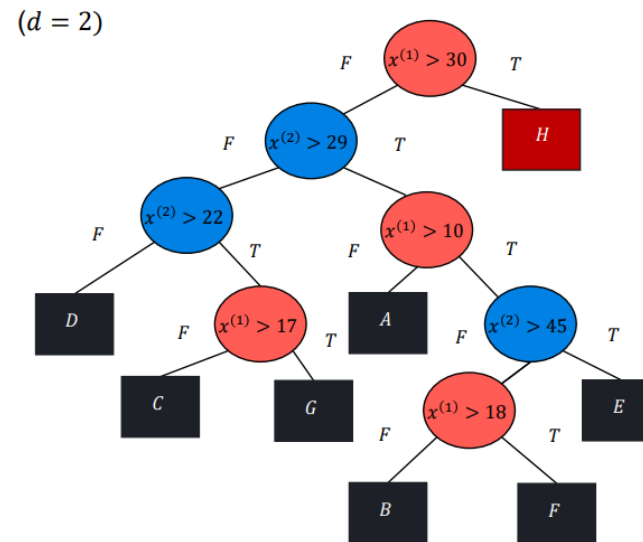
- Intuition: it is easier to separate an object from the other objects by randomly splitting the space



■ Outlier Detection > High-dim > Ensemble-based

□ Isolation forest

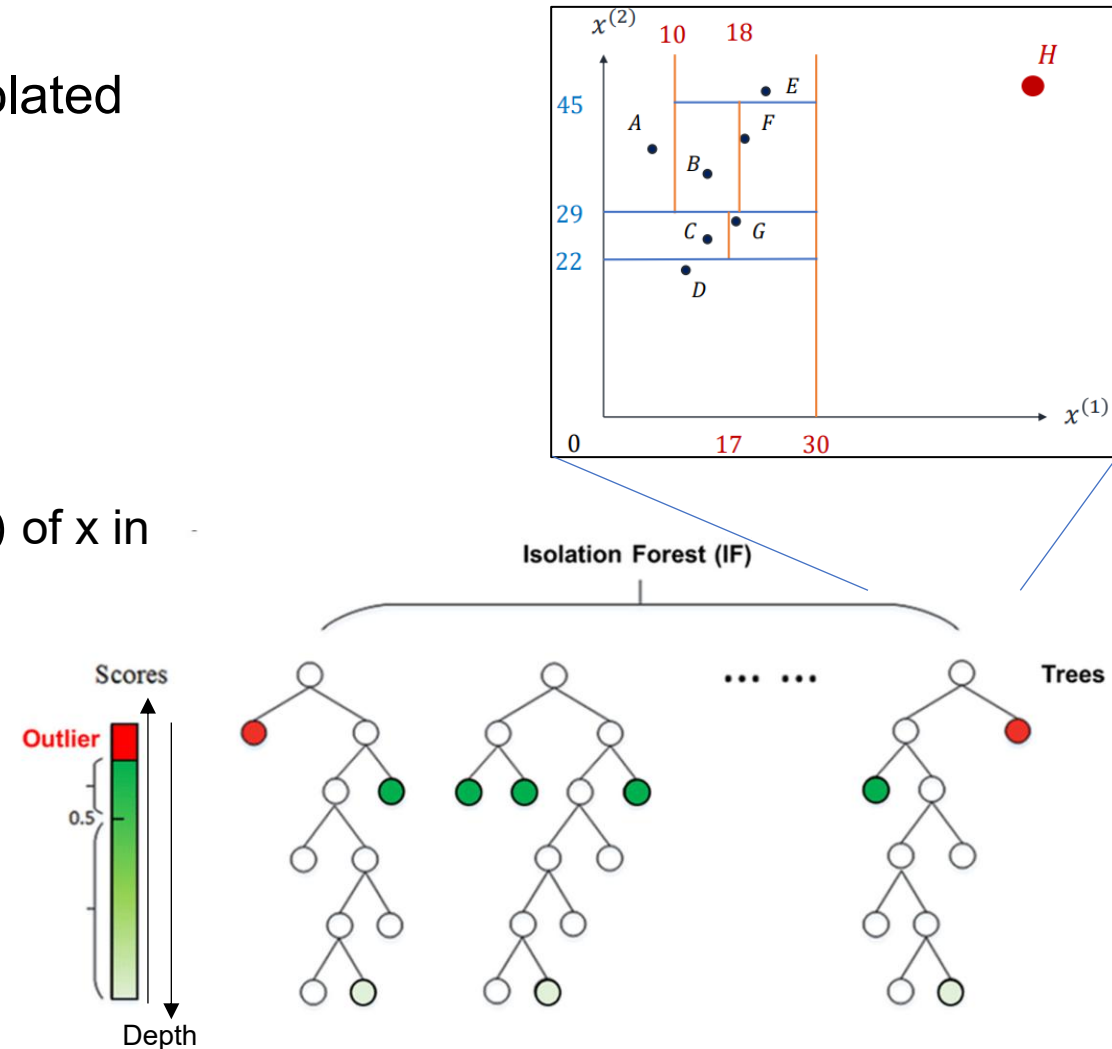
- Build various trees and check when an object is isolated
- Procedure
 - Initialize a forest (i.e., an empty set of trees)
 - For the number of trees t ,
 - Randomly sample a subset of a data set
 - Build a tree from the sampled subset with
 - 1) Randomly selected splitting attribute
 - 2) Randomly selected splitting point
 - Add a tree to the forest



■ Outlier Detection > High-dim > Ensemble-based

□ Isolation forest

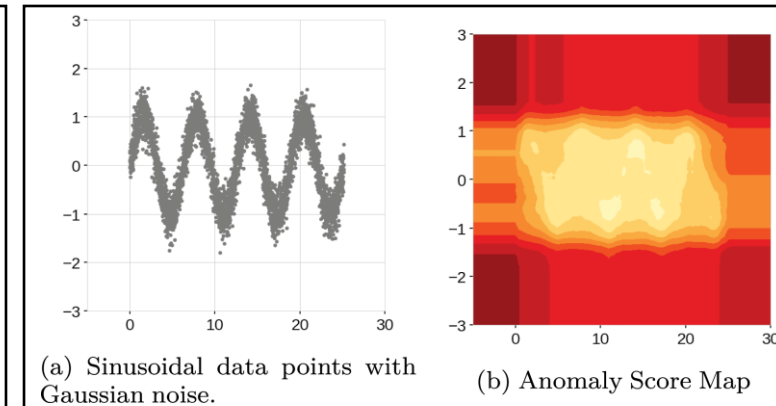
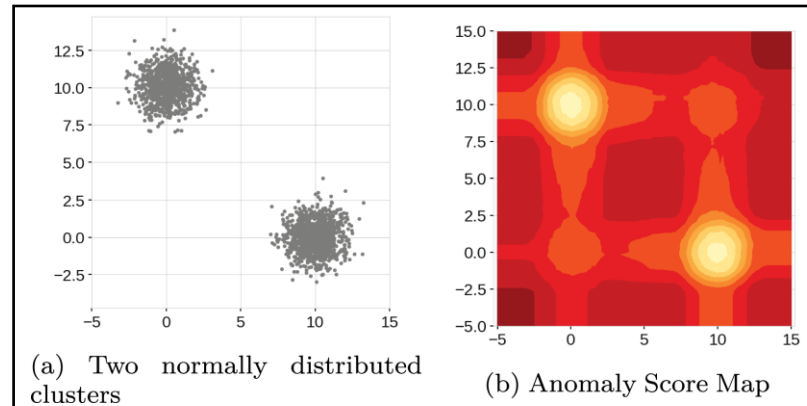
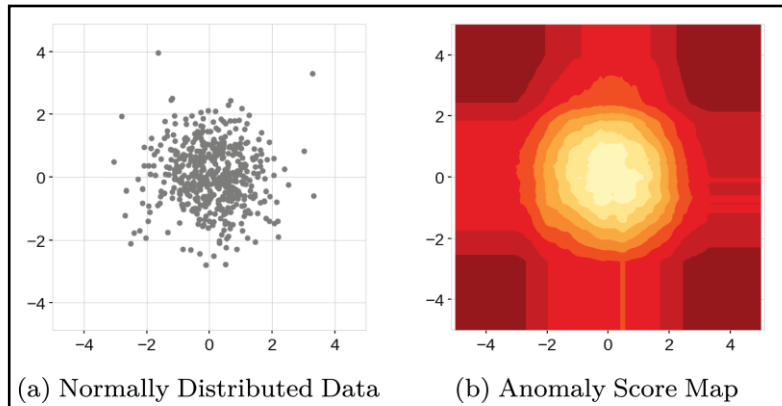
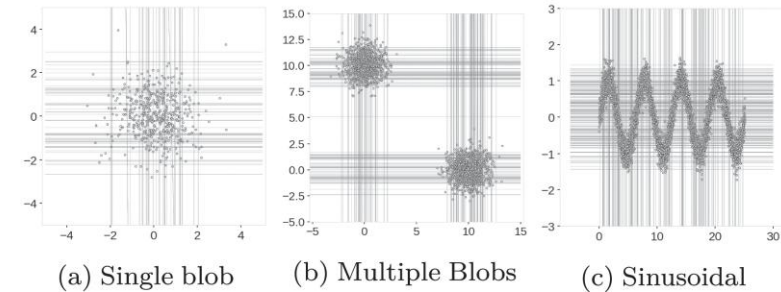
- Build various trees and check when an object is isolated
- Procedure
 - Initialize a forest (i.e., an empty set of trees)
 - Build a set of trees from the sampled subset
- Evaluation
 - For an object x , calculate the average path length $\bar{h}(x)$ of x in each tree from the root node to the external node
 - Calculate the anomaly score $S(x) = 2^{-\bar{h}(x)/L}$
 - L : a normalization constant (i.e., an expected path length)
 - The score $\rightarrow 1$: the path length is small (anomaly)
 - The score $\rightarrow 0.5$: the path length is large (normal)



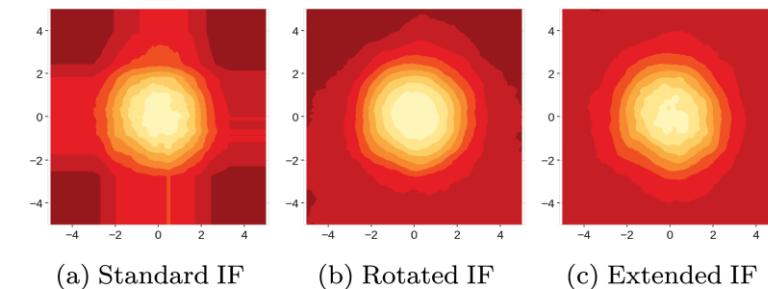
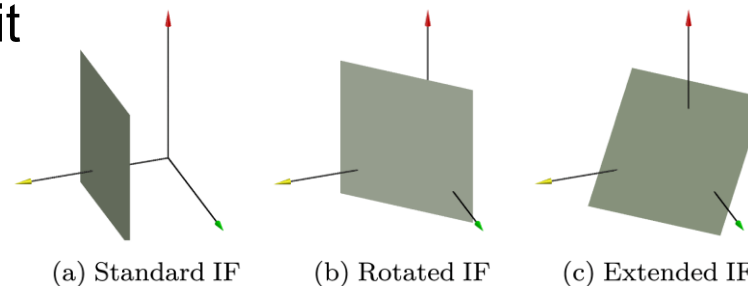
Outlier Detection > High-dim > Ensemble-based

□ Limitation of standard isolation forest

- Ghost area: a region of few objects but lower anomaly scores
 - Owing to random splits for one of the fixed random axes (attributes)
- Example (darker areas indicate higher anomaly scores)



- Extension with hyperplane split
 - Rotated isolation forest
 - Extended isolation forest

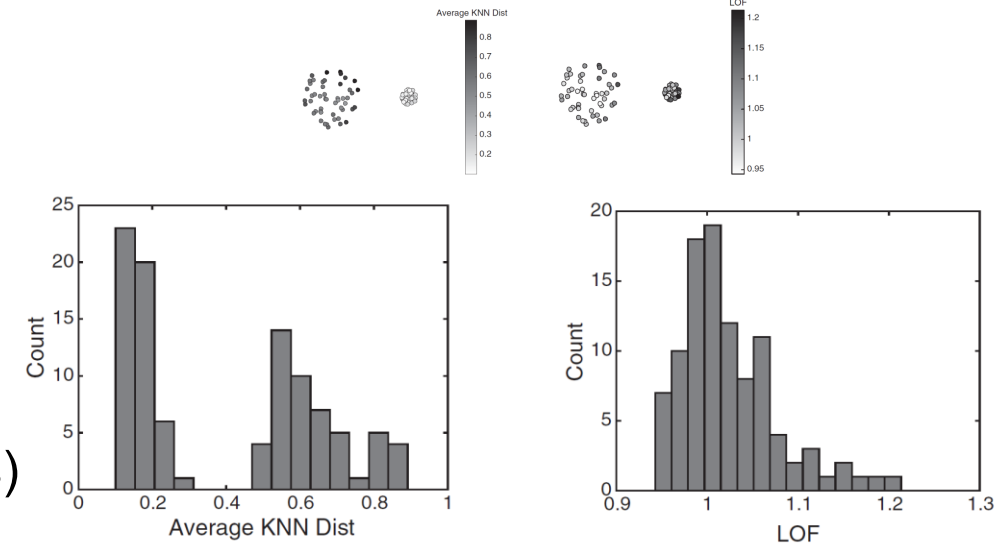


Outline

- ❑ Basic concepts of outlier detection
- ❑ Statistical methods
- ❑ Proximity-based methods
- ❑ Methods for high-dimensional data
- ❑ **Outlier detection evaluation**

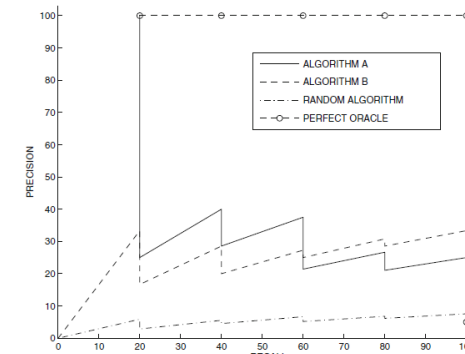
Outlier Detection > Evaluation

- ❑ By definition, outliers are rare
 - The ground-truth labels are not always available
- ❑ When labels are not present
 - Qualitative analysis (e.g., histograms of anomaly scores)
- ❑ When labels are present (i.e., binary classification?)
 - Standard evaluation measures
 - ROC (TPR – FPR) curve → AUC score
 - Precision-Recall(=TPR) curve → Average Precision score
 - More useful when the classes are imbalanced ($P \ll N$)
 - FPR ($FP/(FP+TN)$) in ROC → Precision ($TP/(FP+TP)$)
 - Top-N evaluation measures (N: # of true outliers)
 - Precision@N
$$P@n = \frac{|\{o \in O \mid \text{rank}(o) \leq n\}|}{n}$$
 - Average Precision@N
$$AP = \frac{1}{|O|} \sum_{o \in O} P@ \text{rank}(o)$$

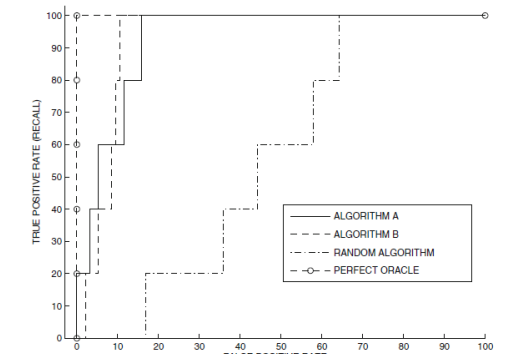


Ranks of true outliers out of 100 objects

Algorithm	Rank of Ground-truth Outliers
Algorithm A	1, 5, 8, 15, 20
Algorithm B	3, 7, 11, 13, 15
Random Algorithm	17, 36, 45, 59, 66
Perfect Oracle	1, 2, 3, 4, 5



(a) Precision-recall



(b) Receiver operating characteristic

Useful Resources

❑ Public repositories

- [Anomaly-detection-resources](#) (a comprehensive collection of relevant resources)
- [PyOD](#) (implementation of most of the popular methods)

❑ Libraries

- [Scikit-learn - outlier detection](#)

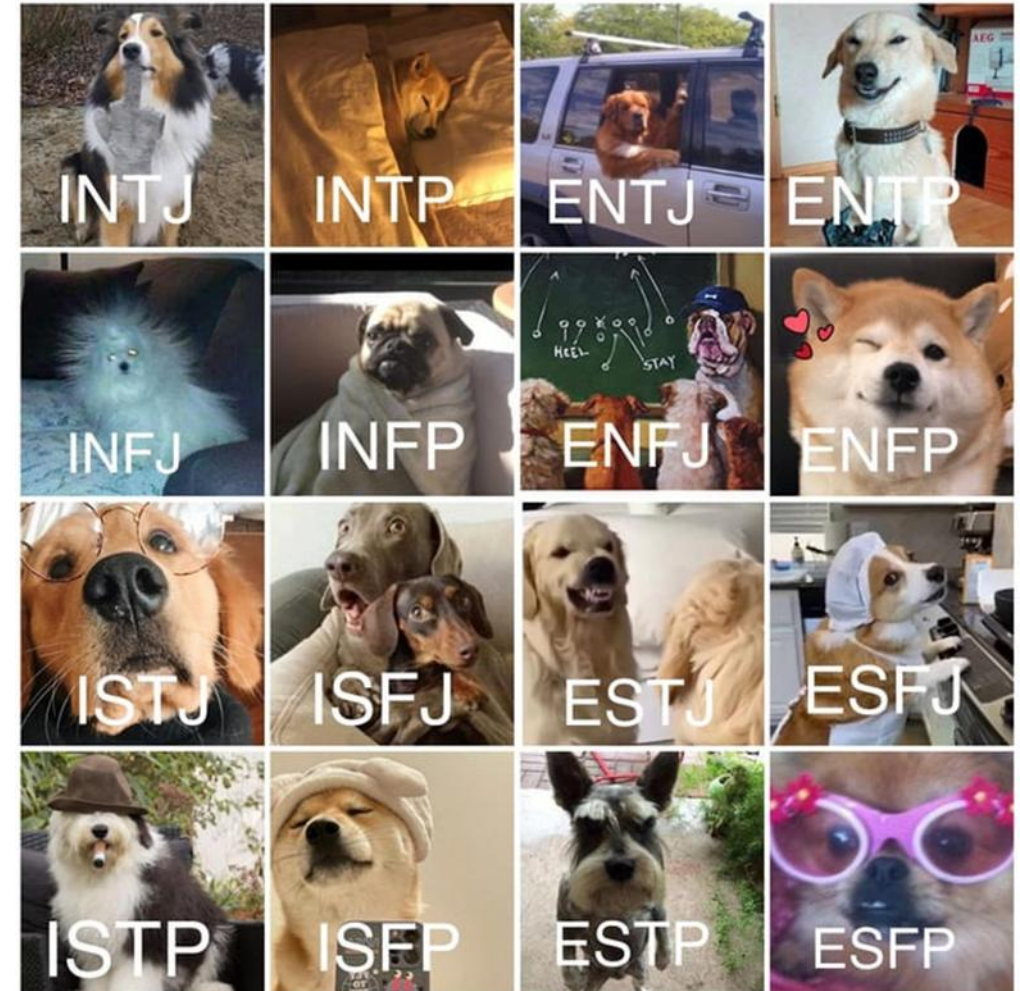
Outline

□ Today

- Basic concepts of outlier detection
- Statistical methods
- Proximity-based methods
- Methods for high-dimensional data
- Outlier detection evaluation

□ Next

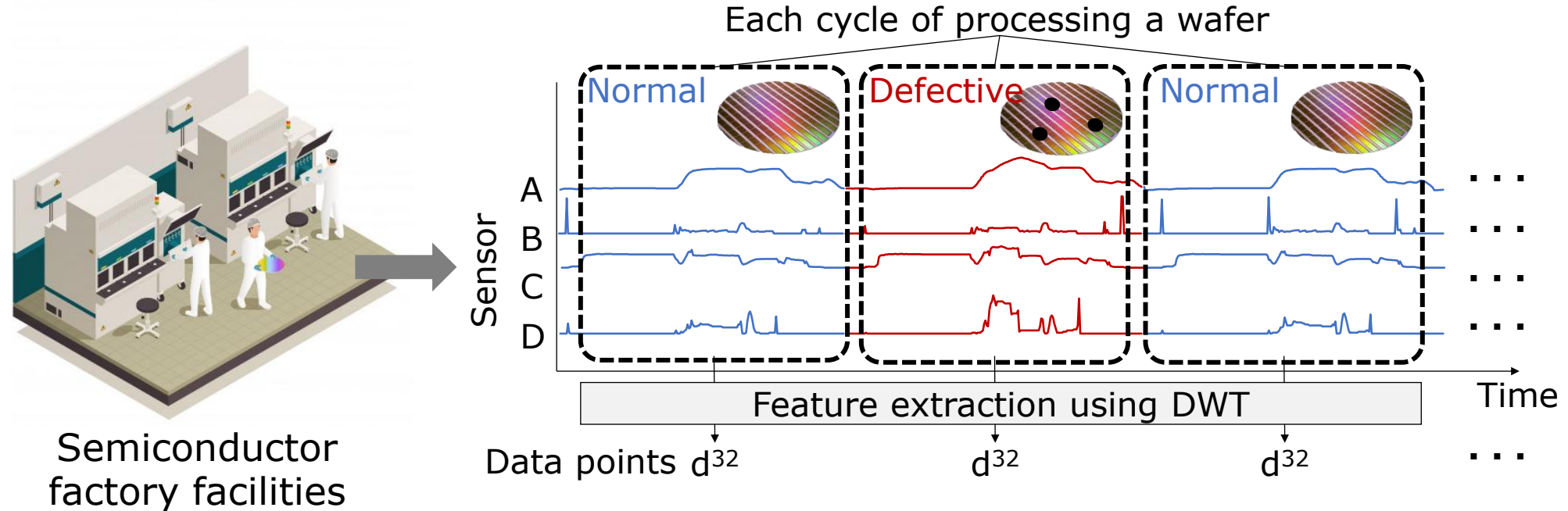
- Classification



[Cookie Slides] Streaming Outlier Detection

KDD 2020 Research

Ultrafast Local Outlier Detection from a Data Stream with Stationary Region Skipping

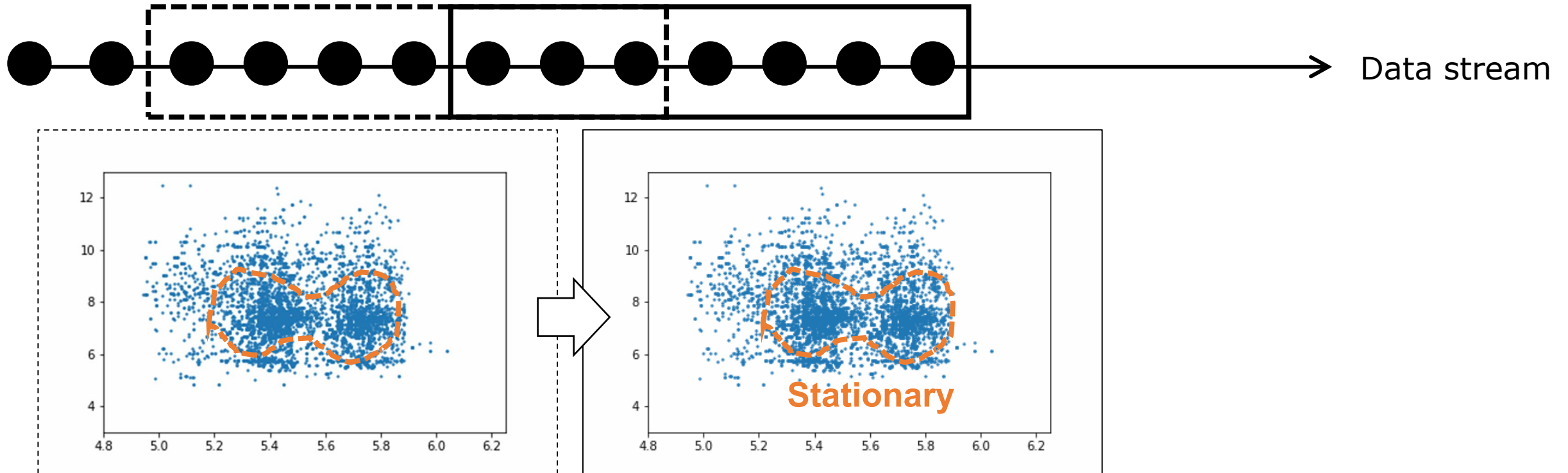


[Cookie Slides] Streaming Outlier Detection

KDD 2020 Research

Ultrafast Local Outlier Detection from a Data Stream with Stationary Region Skipping

Distribution of a real world data stream



[Cookie Slides] Streaming Outlier Detection

KDD 2020 Research

Ultrafast Local Outlier Detection from a Data Stream with Stationary Region Skipping

Data type	Data structure	Data state	Target knowledge	Application
Time-series/ Temporal	Structured	Streaming	Outlier	Fraud detection Fault detection

